

How Much Should Government Pay to Save a Job?*

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Abstract

Governments decide how much to spend preserving jobs using cost-per-job statistics, and a large empirical literature measures what workers lose when jobs are destroyed. Neither number is the social value of the job. A job is a worker-firm match, and the value of preserving it is the joint surplus of the two parties: the worker's welfare loss from separation and the planner-valued firm-side continuation surplus, together with fiscal effects and instrument-specific externalities. Standard supply and demand identify the marginal distributions of worker and firm surplus but not their joint distribution, and a shock's destruction rule and a policy's response rule can select different regions of that joint distribution. This paper develops the framework that prices both margins—the shadow value of averting a policy-marginal separation and the condition under which retention beats worker insurance—and specifies an empirical design for linked employer-employee and firm-accounting data that can estimate the joint distribution of predicted worker and firm surplus and both selection kernels, treating the latent match-level objects as partially identified around those estimates. Social-value adjustments the data do not identify enter as bounds and verdict regions rather than point verdicts.

Keywords: Cost of job loss; Job retention; Separations; Match surplus; Linked employer-employee data; Unemployment insurance

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1 Introduction

The cost of losing a job is among the most intensively measured objects in empirical economics. [Jacobson et al. \[1993\]](#) follow high-tenure Pennsylvania workers displaced in the mass layoffs of the early 1980s and find earnings losses of roughly 25 percent of prior earnings that persist more than five years after separation. [Davis and von Wachter \[2011\]](#) estimate that a high-tenure worker displaced in an expansion loses the present-value equivalent of 1.4 years of pre-displacement earnings, and a worker displaced in a deep recession loses 2.8 years. These estimates anchor how economists and policymakers understand the harm from job destruction: they set the scale for debates over trade policy, place-based policy, and the safety net, and they are the numbers invoked when governments spend to prevent job loss or to compensate it.

A job, however, is a match between a worker and a firm, and the social value of preserving it is the total surplus the match generates, not the worker’s share alone. Worker surplus u is the gap between the value of the job to the worker and the worker’s next-best alternative; firm surplus π is the gap between what the match produces and what it costs the firm. The displacement literature measures the earnings losses that anchor u , and measures them well. Nothing in those research designs measures π , and the designs select on both coordinates at once: tenure restrictions select workers with high u , while mass-layoff and closure conditioning selects firms at their private exit margin—for a closing establishment, private continuation profit at or below zero is close to definitional. Whether the firm-side surplus a planner should count was also low is not definitional; it is an empirical question that no displacement design answers, because private profit and planner-valued surplus can diverge through rents, rigid wages, and organizational capital. The measured worker loss could therefore overstate the joint surplus destroyed, when the destroyed matches sat at firms with little continuation value, or understate it, when firm-side surplus was substantial and the shock destroyed the match anyway. [Bartik \[2015\]](#) argues on worker-side adjustment grounds that measured earnings losses overstate the social cost of lost jobs in benefit-cost

analysis; the wedge studied here is distinct: it sits on the firm side, and the firm-side coordinate has never been measured jointly with the worker’s loss for the population of matches at risk.

The question this paper prices is the policy counterpart of that accounting gap: how much should the government be willing to pay to avert a particular separation? By “save a job” I mean avert the separation of an existing worker-firm match. The object the planner values is the continuation of that match, not employment as a head count. A displaced worker can alternatively be supported through unemployment insurance, wage insurance, transfers, retraining, mobility assistance, or place-based programs; retention is valuable only to the extent that preserving the specific match creates social value relative to those alternatives.

The paper’s organizing device is a triptych. A supply curve locates each worker’s surplus and a demand curve locates each firm’s surplus, so the two familiar diagrams identify the two marginal distributions. But the welfare object depends on how worker and firm surplus are paired across matches—on the joint distribution $G(u, \pi)$, equivalently the copula linking the two margins—and no supply-and-demand diagram identifies the pairing. A worker with much to lose can be matched to a firm with little continuation value, or to a firm whose continuation value is large; the two configurations carry the same worker loss and different social losses. The joint distribution, not either marginal alone, is the paper’s central empirical object.

The second ingredient is selection. A shock destroys particular matches, described by a destruction kernel $p_z(u, \pi)$: the probability that a match at (u, π) separates under shock z . A retention policy saves particular matches, described by a response kernel $q_B(u, \pi)$: the probability that the marginal program dollar averts that match’s separation. The two kernels need not coincide, and neither needs to track total surplus. A shock can destroy one region of the joint distribution while the marginal retention dollar saves another; evaluating the shock’s damage and evaluating the policy’s value are different weighted averages over the same joint distribution.

These objects assemble into a compact welfare rule. The shadow value of the policy-marginal averted separation is the avoided net fiscal cost plus the preserved nonfiscal value—the worker’s welfare loss, the planner-valued firm-side continuation surplus, and any externality the retention instrument mitigates—scaled by the marginal cost of public funds. Retention is worth expanding when this shadow value per dollar of marginal retention cost exceeds the marginal value of the alternative instrument, worker insurance. The distinction between the firm’s private continuation profit, which governs whether the firm separates and whether it takes up a subsidy, and the planner-valued firm surplus, which enters the welfare calculation, runs through every step and does first-order work: the private object is what firm behavior reveals, and the planner object is what policy should price.

The paper’s empirical contribution is a design, stated precisely enough to be implemented or refuted. In a single setting with linked employer-employee and firm-accounting data, the design estimates worker losses, a defensible firm-side continuation-value proxy with an explicit bridge from private to planner-valued surplus, the joint distribution of the two as predicted for every at-risk match, the shock’s destruction kernel over that distribution, and—where the setting contains a retention program with credible variation—the policy-response kernel and the marginal cost per averted separation. Because the match-level coordinates are predicted from pre-shock observables, the design is explicit about what the resulting joint distribution identifies exactly—the kernel-weighted means the welfare rule consumes—and what it only bounds: the latent match-level dependence and any selection operating inside prediction cells. Objects the data do not identify, such as the planner’s weight on measured rents or the differential externalities of retention versus insurance, likewise enter the reporting as bounds and verdict regions: statements of which conclusions survive over the admissible parameter space, rather than point verdicts. [Appendix D](#) reports public-data pilot exercises that demonstrate the accounting is computable at the industry and sector level; the match-level implementation is the paper’s planned next stage, and [Appendix E](#) maps candidate data environments to each estimand.

The paper makes three contributions. First, it develops the two-sided accounting of destroyed match value and the triptych that organizes it, showing that standard supply and demand identify the marginal distributions of worker and firm surplus but not the joint distribution that welfare requires. Second, it derives the shock- and policy-conditioned retention rule and the retention-versus-insurance frontier, in which the value of saving jobs depends on which matches the shock destroys and which matches the marginal program dollar saves. Third, it specifies an implementable linked-data design for the five empirical objects the rule requires, together with a partial-identification framework for reporting welfare conclusions under the adjustments the data leave open. Section 2 builds from the textbook benchmark to the joint distribution; Section 3 states the welfare framework; Section 4 presents the empirical design; Section 5 states the planned outputs of the completed analysis; Section 6 discusses applications; Section 7 concludes. Appendices collect the stylized-case diagrams, formal results and proofs, an illustrative break-even calibration held over from an earlier version of this paper, the public-data pilots, and data-access paths; an online appendix (available from the author) collects the full calibration tables, the evidence review underlying the measurement anchors, and the parametric copula and kernel illustrations.

2 From Supply and Demand to the Joint Distribution

This section builds the paper’s central object in three steps. The first step states the textbook benchmark in which destroyed jobs are the right jobs to destroy, so that job counts carry no welfare content. The second step holds the number of separations fixed and shows that welfare depends on which matches separate, and that the resulting selection loss is measured relative to the efficient assignment, not as the gross surplus of destroyed matches. The third step splits match surplus into its worker and firm coordinates and shows why the joint distribution of the two, rather than either marginal, is the object policy evaluation requires.

2.1 The market-marginal benchmark

Consider a labor market populated by existing worker-firm matches. Each match generates worker surplus u , the value of the job to the worker net of the worker's opportunity cost, and firm surplus π , the match's marginal revenue product net of the wage. Total match surplus $u + \pi$ is the marginal revenue product in excess of the worker's opportunity cost; the wage divides this surplus between the parties but does not change its total. In this section π is the firm's private continuation surplus at the prevailing wage; Section 3 distinguishes it from the planner-valued firm-side surplus. Now let a macroeconomic shock reduce demand. The shock changes the efficient level of employment, so the welfare benchmark is the efficient post-shock allocation, not the pre-shock one: some separations are efficient after demand contracts, and the loss relevant for policy is the surplus destroyed in excess of what the efficient adjustment would have destroyed.

When wages adjust freely, the market clears at a lower wage and lower employment, and the destroyed matches are exactly those whose post-shock total surplus has reached zero at the margin and gone negative inframarginally. Retaining any destroyed match generates a weakly negative welfare gain, and there is no efficiency case for retention. When wages do not adjust, the firm absorbs the shock through layoffs, and employment falls further: some matches with positive total surplus are destroyed because the firm bears the entire shortfall through π . But if separations remain ordered by surplus—the firm sheds its lowest-surplus matches first—the inefficiently destroyed matches are precisely those with the smallest positive surplus, and the welfare loss is a triangle, second order in the wage rigidity.

The content of this benchmark is not that job destruction is small but that it is well selected. In both cases the marginal destroyed match is approximately the right match to destroy, so the number of jobs lost carries almost all the information and the identity of the lost jobs carries almost none. Appendix A reproduces the standard diagrams for both cases. The rest of the paper concerns the situations in which this selection property fails.

2.2 Quantity versus selection

Hold fixed the number of separations a shock forces and ask which matches separate. A sticky wage leaves more workers willing to work than jobs available: every match with positive surplus at the prevailing wage would continue if it could, but only some do. The question is no longer how many matches are destroyed but which ones are rationed out of continuation.

Let $S = u + \pi$ denote total match surplus, let z index the shock, let q_z be the separation rate among at-risk matches, and let D indicate destruction. The efficient rule destroys the lowest-surplus matches first. The actual rule—whatever combination of sectoral exposure, liquidity, credit access, seniority systems, bankruptcy timing, and administrative eligibility governs survival—need not. The welfare loss from misselection, per at-risk match, is

$$q_z \left\{ \mathbb{E}[S \mid D_{\text{actual}}, z] - \mathbb{E}[S \mid D_{\text{efficient}}, z] \right\}, \quad (1)$$

the gap between the average surplus of the matches actually destroyed and the average surplus of the matches the efficient assignment would have destroyed at the same separation rate.

Equation (1) disciplines a claim that is easy to overstate. The gross surplus destroyed by a badly selected shock is a rectangle: q_z matches lost at an average surplus near the population mean. But the gross rectangle is not the welfare loss. The efficient adjustment would itself have destroyed q_z matches, at the bottom of the surplus distribution, and only the excess of the actual loss over that efficient loss is a cost of misselection. When the actual rule is close to surplus-ordered, the excess is small even if the gross rectangle is large; when the actual rule is unrelated to surplus, the excess approaches the gross rectangle less the efficient triangle. Welfare statements in this paper are excess losses relative to the efficient post-shock benchmark throughout.

Literal randomness is not required for the excess loss to be first order. The requirement is

imperfect ordering: the variables that determine survival must be imperfectly rank-correlated with total surplus. That condition is weaker than lottery-like layoffs and is consistent with fully optimizing firms, because the firm’s decision variable is its own surplus, not the total. Direct evidence that separations violate the joint-surplus ordering exists: [Jäger et al. \[2023\]](#) use an Austrian unemployment-insurance extension and its repeal to test whether separations are Coasean-efficient and reject, with matches destroyed by improved worker outside options that efficient bargaining would have preserved. Appendix [A](#) states the one-dimensional formalization, in which the excess loss is expressed through the quantile function of the surplus distribution.

2.3 The triptych

The one-dimensional analysis treats total surplus S as the sorting variable. But separations are decided by parties who each observe their own coordinate: firms respond to π , workers to u , and shocks to neither. Splitting S into (u, π) is therefore not a refinement; it is where the selection problem lives. Two workers with identical surplus can be matched to firms with very different continuation values: firms differ in productivity, market power, intangible capital, and quasi-rents that are only partly passed through to wages, while workers with similar wages differ in tenure, employer-specific human capital, promotion ladders, pension vesting, schedules, commutes, and other amenities. Once the two coordinates are imperfectly aligned, no single supply-and-demand diagram can represent the heterogeneity that matters.

Figure [1](#) makes the point in the geometry of supply and demand itself. The supply panel maps worker rank into worker surplus $u = \bar{w} - s$; the demand panel maps firm rank into post-shock firm surplus $\pi = d_1 - \bar{w}$ after demand falls from d_0 to d_1 at a fixed wage $\bar{w}$; and the joint panel pairs the two coordinates across incumbent matches. The two outer panels are the familiar diagrams, and they identify the two marginal distributions exactly. The joint panel contains the additional information the outer panels cannot supply: which worker is paired with which firm. The figure traces one destroyed match through all three panels: in

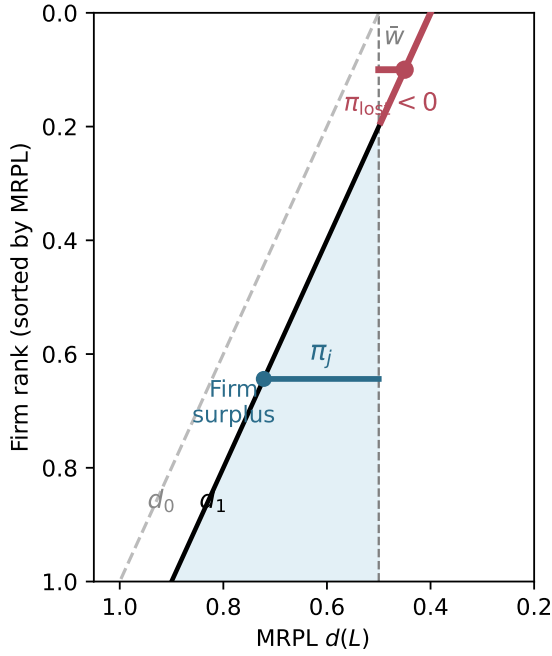
the demand panel its firm surplus is negative, so the firm separates; in the supply panel the same worker’s surplus is positive; in the joint panel the match lies in the band $\pi < 0 < u + \pi$, where the firm’s private rule destroys a match the planner would preserve.

The four regions of the joint panel are descriptive categories, not verdicts. Matches with high u and high π carry the most total surplus; matches with low values of both coordinates sit near the classical margin. Matches with high u and low π must be split by the planner’s frontier: above the $u + \pi = 0$ line, worker surplus outweighs the firm-side shortfall and continuation is efficient; below it, separation is efficient even though the worker’s loss is large. Matches with low u and high π raise a question the diagram cannot settle, namely whether the firm-side value is surplus a planner should count or a rent that merely changes hands; Section 3 takes that question up directly. Attaching policy prescriptions to the quadrants requires the additional structure of Section 3, and the paper resists doing so before then.

Two mechanisms populate the off-diagonal regions with destroyed matches. The first is private decision rules that condition on one coordinate. With rigid wages, incomplete contracts, or failures of within-match bargaining, a firm separates when its own surplus is negative even though total surplus is positive, and a worker quits when $u < 0$ even though the firm-side value is high; [Hall and Lazear \[1984\]](#) and [Bewley \[1999\]](#) document that employment adjustment runs through institutional rules—seniority, divisions, locations—rather than through match-level renegotiation, and the [Jäger et al. \[2023\]](#) rejection of Coasean efficiency shows the resulting band $\pi < 0 < u + \pi$ is not empty. The second mechanism is external destruction: shutdown orders, physical disruption, or sectoral collapse destroy matches by coverage—which sectors, places, and production processes are exposed—without reference to surplus on either coordinate, as when public-health orders closed hospitality in 2020 regardless of the surplus of the matches inside.

The formal object is standard. The joint distribution $G(u, \pi)$ across at-risk matches is determined by its two marginal distributions together with the copula that links them, and the copula—not any parametric family, but the dependence structure itself—is what

Demand (rotated): post-shock d_1 falls below sticky $\bar{w}$



Joint distribution of (u, π)

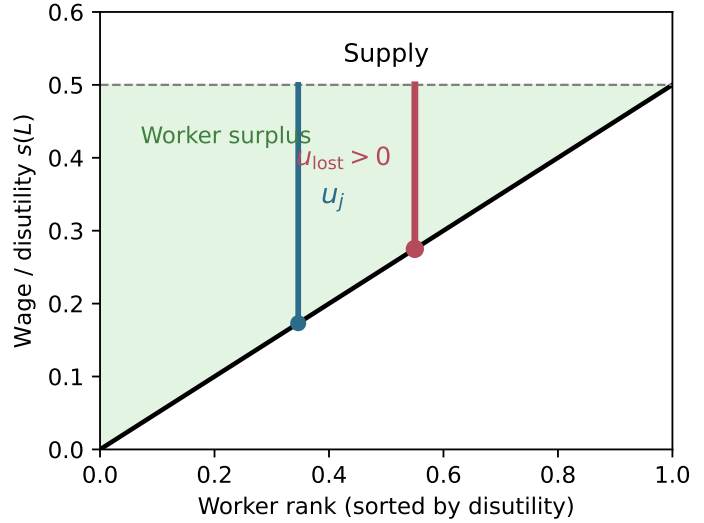
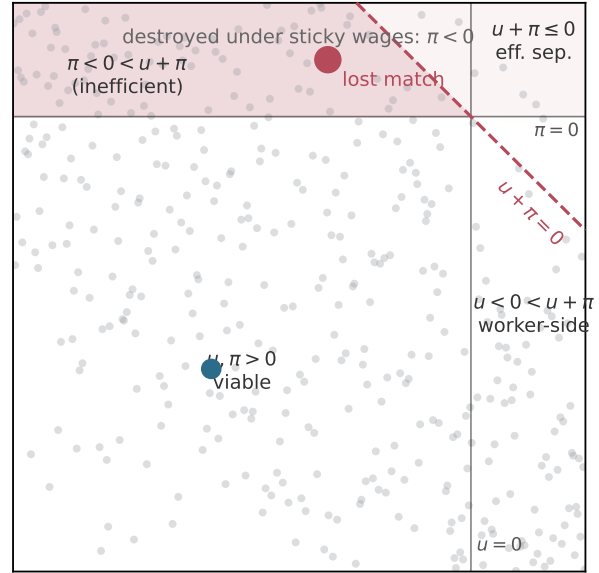


Figure 1: Worker surplus, firm surplus, and their joint distribution. Standard supply and demand identify the two marginal distributions; the welfare object requires the joint.

Note: The supply panel maps worker rank into worker surplus $u = \bar{w} - s$. The demand panel maps firm rank into post-shock firm continuation surplus $\pi = d_1 - \bar{w}$, after demand falls from d_0 (dashed) to d_1 (solid) at the fixed wage $\bar{w}$; the red segment of d_1 marks the destroyed jobs. The red point marks a lost match, with $\pi_{lost} < 0$ in the demand panel and $u_{lost} > 0$ in the supply panel; the blue point marks a surviving inframarginal match j . The joint panel pairs the two coordinates across incumbent matches; total continuation surplus is $u + \pi = d_1 - s$, so the wage cancels. The $\pi = 0$ and $u = 0$ boundaries are the firm's and worker's private cutoffs; the $u + \pi = 0$ frontier through their intersection is the planner's. The lightly shaded strip $\pi < 0$ is selected by the firm-side rule under sticky wages; its match share equals the demand panel's share of destroyed jobs. Within the strip, the darker band $\pi < 0 < u + \pi$, clipped by the planner's frontier, holds matches the firm separates but the planner would retain; the lost match lies there, and separations beyond the frontier are efficient for the planner too. In this figure π is the firm's private continuation surplus at the prevailing wage; Section 3 distinguishes private from planner-valued firm surplus.

the outer panels of the triptych cannot identify. The empirical status of the three pieces is asymmetric: the displacement literature disciplines moments of the marginal distribution of u through the earnings losses it measures, the firm-rent literature supplies moments and candidate proxies for the marginal distribution of π , and the copula for policy-relevant populations of at-risk matches has not been estimated. Making it estimable is the purpose of the design in Section 4.

2.4 From the joint distribution to selection

Welfare evaluation requires one more object beyond G : which matches the event of interest touches. Two kernels do this work. The destruction kernel $p_z(u, \pi)$ gives the probability that a match at (u, π) is destroyed under shock z ; it summarizes the survival-relevant variables—sectoral exposure, liquidity, credit access, wage rigidity, bankruptcy timing—and how they relate to the two surplus coordinates. The response kernel $q_B(u, \pi)$ gives the probability that the marginal dollar of retention spending B averts that match’s separation (Section 3 restores the kernel’s remaining arguments, the insurance environment and the shock, as $q_{B,r,z}$).

The two kernels differ whenever the program’s allocation rule differs from the shock’s. A program that reaches firms through banks favors matches at well-banked firms; a size cutoff favors matches at firms near the threshold; first-come-first-served favors firms with administrative capacity; and take-up requires that the subsidy exceed the firm’s private shortfall, which screens on the private coordinate π^{priv} rather than on total surplus. Each design feature tilts q_B toward a region of G that need not be the region p_z destroys. Pricing the shock’s damage weights G by p_z ; pricing the marginal program dollar weights G by q_B ; the two averages can differ in either direction, and no theoretical argument fixes their ranking.

This section has used no numbers deliberately: which regions particular shocks destroy, and which regions particular programs save, are empirical questions. Parametric illustrations of destruction kernels for familiar shock types, labeled as pedagogy rather than estimates,

are collected in the online appendix. Section 3 embeds G , p_z , and q_B in the welfare rule, and Section 4 is a design for estimating all three.

3 Welfare and Policy Framework

This section states the planner’s problem once, in the objects the rest of the paper estimates. Section 3.1 defines the primitives and their units. Section 3.2 defines the shock-conditioned and policy-conditioned values of a preserved match. Section 3.3 derives the shadow value and the retention condition. Section 3.4 solves the joint retention-insurance problem and states the frontier that organizes the paper’s welfare reporting. Section 3.5 derives the private implementation rule and the partial-identification thresholds. Formal statements and proofs are in Appendix B.

3.1 Objects, units, and counterfactuals

Index at-risk matches by i . The insurance environment r —unemployment insurance generosity, wage insurance, transfers—is a policy variable throughout, and worker-side objects are defined conditional on it.

The worker-side object $u_i(r)$ is the present-value welfare loss the worker suffers if match i separates, evaluated under insurance environment r . It is not the gross earnings loss: it nets out taxes and transfers, the insurance payments the worker receives under r , the value of reemployment along the worker’s actual reallocation path, and the nonpecuniary amenity value of the lost job, and it accounts for consumption smoothing. Because reemployment enters the counterfactual, worker-side reallocation is inside $u_i(r)$ by construction; no separate reallocation adjustment appears in the framework, and adding one would double count.

The firm-side object comes in two versions, and the distinction carries much of the paper’s content. Private continuation profit π_i^{priv} is the firm’s own net value of continuing match i at prevailing terms; it governs the firm’s separation decision and its willingness to take

up a subsidy. Planner-valued firm-side surplus π_i^S is the portion of firm-side continuation value that the planner counts as social surplus. The two diverge through product-market rents that would transfer to competitors rather than disappear, monopsony rents, tax-driven components, and organizational or match-specific capital not capitalized into either profits or wages. A no-double-counting rule governs the split between u and π^S : firm-side value that passes through to the worker as premia, amenities, or match-specific wage growth is already embedded in the worker’s earnings-loss counterfactual and belongs in u , not π^S . Like u , the firm-side counterfactual is net of reallocation: π^S is continuation value relative to the redeployment of the firm’s capital and the refilling of its vacancies, not relative to zero.

The remaining objects are the policy environment. The externality terms X_i^R and X_i^I are the external effects mitigated by retention and by insurance or place-based support respectively—search congestion, local multipliers, agglomeration, scarring spillovers, fiscal stress in affected places—and they are instrument-specific: nothing requires that saving the match and insuring the worker mitigate the same externalities. The fiscal term $\Phi_i(r)$ is the net fiscal cost avoided when the separation is averted: unemployment insurance and transfer outlays not paid, payroll and income tax revenue not lost, evaluated under r and consistent with the netting inside $u_i(r)$, so that an insurance payment counted as a cost in Φ is counted as a receipt in u . The marginal cost of public funds is $\lambda \geq 1$. The retention technology is summarized by $C(B)$, the marginal public cost per averted separation at program spending B ; where a program’s cost is reported per retained job-year rather than per averted separation, the conversion is a substantive modeling step, and [Appendix C](#) shows how the choice moves conclusions. All match-level objects are present values over the remaining life of the match, so shock persistence enters through the continued-match counterfactual: a persistent shock lowers the remaining continuation value the match would generate if preserved, while lengthening the horizon over which a bridge subsidy must be paid; it need not shorten the duration of the worker’s losses, which can rise as reemployment prospects deteriorate.

3.2 Shock-conditioned and policy-conditioned value

Let $G(u, \pi^{\text{priv}})$ denote the joint distribution of worker loss and private firm surplus among matches at risk before the shock. The private coordinate is deliberate: separations, subsidy take-up, and the measurement of Section 4 all operate on π^{priv} , so G is the object that behavior and data discipline. A planner bridge carries it to the welfare coordinate, $\pi^S = m(\pi^{\text{priv}})$ —in the baseline the scalar reweighting $m(\pi^{\text{priv}}) = \omega \pi^{\text{priv}}$, with separately measured components added where the data support them—and G^S denotes the induced distribution of (u, π^S) . Because a scalar $\omega > 0$ is monotone, the bridge preserves ranks, the copula, and every conditioning set, so dependence and kernels estimated on the measured coordinate carry to the welfare coordinate by reweighting alone; richer bridges—component-wise treatment of product-market rents, redeployable capital, or organizational capital—do not preserve ranks, and they enter only through the sensitivity reporting of Section 4.7, not through the point design. The destruction kernel is

$$p_z(u, \pi^{\text{priv}}) = \Pr(D = 1 \mid u, \pi^{\text{priv}}, z), \quad (2)$$

the probability a match at (u, π^{priv}) is destroyed under shock z , and $q_{B,r,z}(u, \pi^{\text{priv}})$ is the policy-marginal response kernel: the probability that the marginal dollar of retention spending averts the match's separation, at spending B , insurance environment r , and shock z . Both kernels are statistical objects: they record destruction and rescue probabilities by location in G , whatever structural rule—private decisions on π^{priv} , program allocation, sectoral coverage—generates them. Two weighted means then price the two events of interest,

$$\bar{S}^D(z) = \frac{\mathbb{E}[(u + \pi^S) p_z]}{\mathbb{E}[p_z]}, \quad \bar{S}^m(B, r, z) = \frac{\mathbb{E}[(u + \pi^S) q_{B,r,z}]}{\mathbb{E}[q_{B,r,z}]}, \quad (3)$$

with expectations over G and $\pi^S = m(\pi^{\text{priv}})$ under the bridge. The first is the average total surplus of the matches the shock destroys: it prices the shock's damage and is the object that

a complete accounting of a destruction episode should report. The second is the average total surplus of the matches the marginal program dollar saves: it is the policy object, the value side of the retention decision. The two coincide only when the program saves a representative draw of what the shock destroys, $q_{B,r,z} \propto p_z$, and Section 2.4 explained why program design generically breaks that proportionality. Appendix B states the covariance identity behind both means—each equals the unconditional mean plus a covariance term between surplus and its kernel—and the conditions under which conditioning reverses cross-cell policy rankings.

3.3 The shadow value and the retention condition

Consider a marginal increase in retention spending. It averts separations at marginal cost $C(B)$ each; each averted separation preserves nonfiscal value $u(r) + \pi^S + X^R$ on average among the saved matches and avoids fiscal cost $\Phi(r)$; and each public dollar costs λ in welfare units. Writing $\bar{u}^m(r)$, $\bar{\pi}^{S,m}$, $\bar{X}^{R,m}$, and $\bar{\Phi}^m(r)$ for the averages over the policy-marginal saved matches—the $q_{B,r,z}$ -weighted means of equation (3)—the shadow value of the policy-marginal averted separation is

$$P^m(B, r, z) = \bar{\Phi}^m(r) + \frac{\bar{u}^m(r) + \bar{\pi}^{S,m} + \bar{X}^{R,m}}{\lambda}. \quad (4)$$

For a tax-financed planner, retention spending should expand while $P^m/C > 1$ and stop where $P^m(B^*, r, z) = C(B^*)$ at an active interior margin; cells where $P^m/C \leq 1$ at $B = 0$ get no retention spending. The classical benchmark of Section 2.1 is the evaluation of (4) at $\bar{u}^m + \bar{\pi}^{S,m} \approx 0$: when the saved matches are market-marginal, the shadow value collapses to the avoided fiscal cost, and expensive retention cannot clear the bar. Duration enters both sides of the condition: persistence of the shock lowers the remaining continuation value behind $\bar{u}^m$ and $\bar{\pi}^{S,m}$ while extending the period over which the program must keep paying, so the same instrument can clear the bar under a temporary shock and fail it under a persistent one.

3.4 Retention versus insurance

The relevant alternative to saving a job is rarely doing nothing; it is insuring the worker after separation. The two instruments draw on the same public funds and act on the same displaced pool, so the optimum is a joint object. A one-cell version makes the structure transparent. A shock would destroy D matches; the planner chooses the number J to save through retention, at spending $B(J)$ with rising marginal cost $C(J) = B'(J)$, and the insurance generosity r received by the $D - J$ workers displaced anyway. Let u_0 be the gross worker loss absent insurance, $h(r)$ the social value of generosity r to a displaced worker, inclusive of any insurance-side externality X^I (with $h' > 0$, $h'' < 0$), $K(r)$ the fiscal cost of providing it, ϕ the other fiscal cost of a separation, and $A = \pi^S + X^R$ the non-worker continuation value. Welfare is

$$W(J, r) = J(A + u_0 + \lambda\phi) + (D - J)[h(r) - \lambda K(r)] - \lambda B(J). \quad (5)$$

The retention first-order condition is $C(J^*) = P^m(r^*)$ with $u(r) = u_0 - h(r)$ and $\Phi(r) = \phi + K(r)$: insurance lowers the worker-loss reason to retain and raises the avoided-fiscal-transfer reason, and at the insurance optimum the two channels exactly offset locally. The insurance first-order condition is the Baily-Chetty condition: the marginal welfare value of a public dollar of insurance,

$$\Gamma(r) \equiv \frac{h'(r)}{\lambda K'(r)},$$

equals one at an interior insurance optimum.

The financing environment determines the comparison the retention margin must win. A tax-financed planner expands each instrument to its own margin: retention to $P^m/C = 1$ and insurance to $\Gamma = 1$, each condition holding independently. A planner allocating a fixed stabilization budget between the two instruments instead equalizes marginal welfare per

dollar across them: retention is active in a cell only where

$$\frac{P^m(B, r, z)}{C(B)} \geq \Gamma(B, r, z), \quad (6)$$

with equality at an interior allocation. Equation (6) is the paper’s organizing welfare statement, and it should be read as a frontier rather than a number: it partitions the space of empirical estimates and unidentified adjustments into regions where retention dominates, regions where insurance dominates, and a boundary whose location the data discipline. Because the externality terms are instrument-specific, the comparison naturally admits that insurance and place-based support mitigate different externalities than retention does; a complete comparison prices X^R against X^I rather than assuming either is zero, as the one-cell problem does through A and h respectively. When retention is unavailable in every cell, only the insurance condition remains and the framework collapses to Baily [1978] and Chetty [2006]; the joint problem nests the Baily-Chetty program and extends it to the case in which preserving matches is feasible and match surplus is first order. Appendix B states the general many-cell problem, its Kuhn-Tucker conditions, and the proof.

Holding insurance fixed at $\bar{r}$, condition (6) rearranges to a threshold on the marginal cost per saved job,

$$C^* = \frac{\bar{u}^m(\bar{r}) + \bar{\pi}^{S,m} + \bar{X}^{R,m} + \lambda \bar{\Phi}^m(\bar{r})}{\lambda \Gamma(\bar{r})}, \quad (7)$$

with retention preferred where the program’s marginal cost falls below C^* . The threshold rises with everything worth preserving and falls as insurance becomes more valuable. It is the natural summary statistic for empirical work because its ingredients separate cleanly into estimated, computed, and bounded components, which is how Section 4 organizes measurement.

3.5 Private implementation and partial identification

The firm's behavior determines what retention costs. A firm continues match i if and only if $\pi_i^{\text{priv}} + b_i \geq 0$, where b_i is a per-match subsidy, so the minimum subsidy that preserves a threatened match is

$$b_i^{\min} = \max\{0, -\pi_i^{\text{priv}}\}. \quad (8)$$

The planner should pay it only when the social value of preservation exceeds the private shortfall plus program delivery cost: the firm separates whenever its private surplus is negative, and the planner bridges the gap only for matches whose total planner-valued surplus warrants it. The private coordinate thus enters the framework twice, once through take-up—subsidies below $b_i^{\min}$ save nothing—and once through cost, since $b_i^{\min}$ paid over the shock's duration is the floor on what preservation can cost.

Inverting the frontier isolates the object most likely to remain unidentified. Fixing the estimated worker loss, fiscal term, and cost, condition (6) holds if and only if the planner-valued firm surplus of the saved matches exceeds

$$\pi_{\text{rescue}}^S = \lambda(\Gamma C - \bar{\Phi}^m) - \bar{u}^m - \bar{X}^{R,m}, \quad (9)$$

with $\Gamma = 1$ in the tax-financed case. Equation (9) converts the retention question into a statement a reader can evaluate: given what the data say about worker losses, fiscal flows, and costs, how much planner-valued firm-side surplus would the saved matches need to carry? The completed empirical analysis reports its welfare conclusions in this inverted form, as verdict regions over the objects that remain unidentified after estimation: the planner's weight on measured private operating surplus, the remaining duration of the match, the marginal retention cost, and the differential externality $X^R - X^I$. Conclusions that hold across the admissible region are reported as findings; conclusions that flip within it are reported as frontiers. Section 4 turns to what the data can pin down.

4 Empirical Strategy

This section specifies the empirical design at the level of estimands, required variation, and planned outputs, so that a reader with access to a candidate data environment can judge feasibility directly. Table 1 summarizes the design in one place: each row is one of the five empirical objects, its empirical counterpart, the data and variation that identify it, and what remains assumed after estimation.

4.1 Empirical targets

The framework requires five objects, and they nest. The first is the worker-side loss u_i . The second is a firm-side value proxy: the private continuation object $\hat{\pi}_i^{\text{priv}}$ that governs separation and take-up, and the bridge from it to planner-valued surplus $\hat{\pi}_i^S$, estimated or bounded. The third is the joint distribution G of the two among matches at risk before the shock, on the measured coordinate of Section 3.2. The fourth is the destruction kernel $p_z(u, \pi)$. The fifth, available only in settings that contain a retention program with credible variation, is the response kernel $q_{B,r,z}(u, \pi)$ together with the marginal cost per averted separation C .

The nesting defines two tiers of result. Objects one through four deliver the complete accounting of a destruction episode: the joint distribution of what was at risk, which region of it the shock destroyed, and the destroyed surplus $\bar{S}^D(z)$ against its efficient-selection benchmark—the empirical content of equation (1). That tier requires no retention policy and answers the question the displacement literature’s estimates are commonly used to inform but cannot, by themselves, answer. The fifth object upgrades the accounting to a causal policy evaluation: what the marginal retention dollar saved, at what cost, and how the saved region compares with the destroyed region. A setting without a retention program still yields the rescue threshold (9) and the cost threshold (7) as statements about what any hypothetical program would have to achieve.

4.2 Measuring the worker-side loss

The worker-side design is the displacement literature’s, applied inside the linked panel and extended in two directions. Separation and the comparison group are defined before outcomes are observed: separators are workers whose match ends during the shock window at establishments meeting an exposure or event definition fixed ex ante, and the comparison group is otherwise similar workers—matched on pre-shock earnings trajectory, tenure, industry, and demographics—whose matches survive, including workers at exposed establishments whose matches survive, so that the design can distinguish the effect of separation from the effect of exposure. Dynamic earnings effects come from a standard matched event-study specification, cumulated to a present value over a stated horizon at a stated discount rate [Jacobson et al., 1993, Davis and von Wachter, 2011, Lachowska et al., 2020, Schmieder et al., 2023].

The first extension converts earnings to welfare. The estimand $u_i(r)$ is a welfare loss under the prevailing insurance environment, so the earnings path is adjusted for taxes and transfers, observed unemployment-insurance receipt, and—where consumption or expenditure data permit—consumption smoothing; the nonpecuniary amenity value of the lost job is not observed and is carried as a bounded adjustment rather than a point estimate. The paper reports both the reduced-form present-value earnings loss, which is comparable to the literature, and the welfare-converted interval that enters the framework.

The second extension makes the loss predictable ex ante. The joint distribution of Section 4.4 requires a match-level $\hat{u}_i$ defined for at-risk matches before destruction is realized, so heterogeneous losses are estimated as a function of pre-shock observables only—tenure, age, earnings level and trajectory, industry, local labor market, firm characteristics—and the fitted function is used to score every at-risk match. Conditioning only on pre-shock information is what prevents the joint distribution from being built on post-treatment outcomes.

4.3 Measuring firm-side continuation value

The empirical unit is the worker-establishment match where the data permit, and the establishment-by-worker-cell otherwise. The core measurement is a pre-shock operating surplus: value added (or revenue less intermediate inputs) less the wage bill less a rental charge on capital, per worker, at the establishment level—the quasi-rent that continuation preserves. Three refinements separate this accounting object from the theoretical coordinate. First, establishment fixed costs must be separated from marginal match value: the average quasi-rent per worker overstates the value of the marginal match when fixed costs are large, so the design reports both average and estimated marginal versions and carries the gap as a bound. Second, forward-looking outcomes—survival, subsequent revenue and employment growth, productivity, market value where listed—are used to validate the pre-shock proxy, not to construct it; a proxy built from future outcomes would mechanically encode the shock it is meant to predict. Third, the mapping from establishment-level surplus to match-level π_i is an estimand, not an identity: within-establishment allocation is disciplined by observable match characteristics (occupation, task, tenure complementarity) where possible and otherwise bounded between equal-division and marginal-product allocations.

The bridge from private to planner-valued surplus—the map m of Section 3.2—is parameterized rather than assumed away. Measured quasi-rents embed product-market rents that would partially transfer to competitors rather than vanish, monopsony rents, and tax components, alongside organizational and match capital that is genuine social surplus; the design expresses $\hat{\pi}_i^S = \omega \hat{\pi}_i^{\text{priv}} + (\text{components measured separately})$ and treats the planner weight ω as a reporting axis in the verdict regions of Section 4.7 rather than as a calibrated constant. The firm-rent literature provides the anchors for what the components can plausibly be [Card et al., 2013, Kline et al., 2019, Sorkin, 2018].

4.4 Estimating the joint distribution

The at-risk population is defined before the shock: all matches active at exposed and comparison establishments at a baseline date. Each match is scored with predicted worker loss $\hat{u}_i = \mathbb{E}[u_i \mid X_i]$ and predicted firm-side value $\hat{\pi}_i$, both functions of pre-shock observables X_i only. The primary estimand is G_X , the joint distribution of the two scores across at-risk matches. G_X is not the latent match-level distribution G : the worker score is a conditional mean and the firm score an allocated establishment-level proxy, so G_X compresses the residual heterogeneity within prediction cells— $u_i - \mathbb{E}[u_i \mid X_i]$, and the match-level deviation of π_i from its allocated value—and no reliability correction recovers it. The distinction costs less than it might. The welfare rule of Section 3 consumes kernel-weighted means, and for any kernel that is itself a function of the observables, iterated expectations makes the score-based mean equal the latent mean exactly: $\mathbb{E}[(u + \pi)k(X)]/\mathbb{E}[k(X)]$ is identified from G_X alone. What G_X does not identify is the latent copula and quadrant masses, which the score joint attenuates, and any selection operating within prediction cells—and the latter is first order here, because the firm’s separation rule conditions on its own π^{priv} , not on the econometrician’s prediction of it. The design therefore reports G_X as the estimand, treats the latent objects as set-identified around it, and carries the within-cell residual as an explicit sensitivity axis; formally this is the score-level analog of the observability decomposition of Appendix B.3, with prediction cells in the role of public-data cells. The estimated firm-side coordinate is the private proxy $\hat{\pi}_i^{\text{priv}}$ —the measured object; the planner bridge of Section 4.3 rescales welfare integrals to the π^S coordinate downstream, so the copula and kernels are estimated once on the measured coordinate and reweighted under each planner weight ω . The shorthand $\hat{\pi}_i$ refers to this measured score throughout. Because both scores are generated regressors, they are estimated with cross-fitting: the prediction functions are fit on one part of the sample and applied to another, so that estimation noise common to both coordinates does not manufacture dependence [Chernozhukov et al., 2018]. The marginals and the copula of G_X are estimated nonparametrically—empirical marginals and the empirical copula of

the score ranks—at the sample sizes linked administrative data provide; parametric copulas enter only as smoothed robustness checks, not as the estimator.

The reported objects are the ones the theory uses, computed on G_X and flagged as such. The rank correlation and the copula summarize sorting; the four quadrant masses of Figure 1, with the planner’s frontier drawn in, summarize where the at-risk population sits; and a within- versus between-establishment decomposition reports how much of the joint variation public aggregates could ever capture, the empirical counterpart of the observability decomposition in Appendix B. The headline exhibit is the empirical triptych: the two estimated marginal distributions in the outer panels and the estimated joint density in the corner panel, replacing the stylized geometry of Figure 1 with data. Estimation noise in either score attenuates the estimated dependence beyond the compression already discussed, so the design reports attenuation diagnostics from split-sample reliability ratios and presents the copula results as a range between uncorrected and reliability-corrected estimates; the reliability correction addresses estimation noise only, and the residual within-cell heterogeneity it cannot address is what the sensitivity axis of Section 4.7 carries.

4.5 Identifying shock selection

Identification of p_z requires a shock whose incidence is credible: exposure that is predetermined with respect to match surplus, such as industry-level import competition instrumented by other-country imports, credit-supply variation through pre-crisis banking relationships, or administrative shutdown rules [Acemoglu et al., 2016, Chodorow-Reich, 2014]. The kernel is estimated as the separation probability over the pre-shock score distribution: a flexible model of destruction on $(\hat{u}_i, \hat{\pi}_i)$, exposure, and their interactions, with the fitted surface $\hat{p}_z(\hat{u}, \hat{\pi})$ reported as a heat map over $\hat{G}_X$. The fitted surface is the projection of destruction onto the score plane—the component of the kernel that observables carry. Because firms separate on their own π^{priv} , destruction can also select within prediction cells; that within-cell component is invisible to any estimator that conditions on observables, and the reporting

bounds it through the sensitivity axis of Section 4.7 rather than treating the projection as complete. The projected surface, not any two-sector comparison, is the headline selection result: it shows directly which region of the score distribution the shock destroyed.

Two threats get explicit designs. Endogenous exit—firms or workers who would have separated anyway—is addressed by event-study pretrend tests on both worker outcomes and establishment outcomes, and by distinguishing destruction at exposed establishments from destruction among exposed matches within surviving establishments. And because $\hat{p}_z$ conditions on generated scores, the same cross-fitting discipline applies as in Section 4.4. With the kernel in hand, the accounting deliverables follow: the selection-weighted destroyed surplus $\bar{S}^D(z)$ —whose between-cell component is identified exactly by the iterated-expectations argument of Section 4.4, and whose within-cell covariance between surplus and destruction is bounded, not estimated—its efficient-bottom- q_z benchmark, and the excess selection loss of equation (1), each with uncertainty propagated from all estimation stages. The efficient benchmark is computed on score-total-surplus, whose quantiles are compressed relative to the latent quantiles, and is flagged accordingly: the score-based excess loss understates selection operating within prediction cells.

4.6 Identifying the policy-response kernel and cost

Where the setting contains a retention program, the design adds the causal tier. The program’s rules are documented first: eligibility, subsidy amount and duration, application timing, and delivery channel. Identification of $q_{B,r,z}$ requires variation in treatment among comparable threatened matches: eligibility discontinuities at size or revenue cutoffs, lender- or administrator-driven variation in access, randomized or staggered rollout, or oversubscription lotteries. The estimand is not the average effect of receipt but the marginal saved match: which threatened matches the last program dollar preserved. Complier characterization delivers an instrument-local version of this object—the distribution of $(\hat{u}_i, \hat{\pi}_i)$ among matches whose survival the instrument moved, valid under exclusion and monotonicity [Imbens and

Angrist, 1994]—and plotting it over the same pre-shock joint distribution as $\hat{p}_z$ makes the re-targeting question visual and testable: does the program save the region the shock destroys, or a different one? Reading the complier distribution as the marginal-dollar kernel $q_{B,r,z}$ is a further step: it requires that the instrument’s compliers stand in for the matches the next program dollar would reach, the policy-relevant treatment-effect condition of Heckman and Vytlacil [2005], and the reporting states which instruments in the setting plausibly satisfy it. The cost side is the marginal public cost per averted separation, from the same variation, in the units of P^m ; where the program pays per job-year, the conversion to per-separation units is reported as its own estimate with its own uncertainty rather than absorbed into an assumed scalar.

Where the setting contains no retention program, the paper says so plainly and reports the accounting tier plus thresholds: the estimated $\bar{S}^D(z)$ and joint distribution imply the cost threshold C^* below which a retention instrument would have dominated insurance, and the rescue threshold π_{rescue}^S the saved matches would have needed to carry. Those are statements about the policy space, identified without any policy.

4.7 From estimates to partial welfare results

The welfare reporting is pre-specified to avoid the false precision the framework itself warns against. Four outputs are planned. First, the estimated joint distribution with uncertainty: marginals, copula, quadrant masses, sorting statistics. Second, the two selection-weighted values, $\bar{S}^D(z)$ for the shock and $\bar{S}^m(B, r, z)$ where a program exists, against the efficient-selection benchmark. Third, the thresholds: the firm-side rescue value (9) and the retention-cost threshold (7), with estimated ingredients substituted and remaining ingredients left explicit. Fourth, retention-versus-insurance verdict regions: for each cell or match stratum, the region of the unidentified-parameter space in which retention dominates, drawn over at least five axes—the planner weight ω on measured private operating surplus, remaining match duration or shock persistence, the marginal retention cost C , the differential external-

ity $X^R - X^I$, and the within-cell residual: the share of dependence and of selection operating inside prediction cells, which the score-based estimates cannot see. The reported statistic is the share of matches or cells for which the verdict is robust across the entire admissible region, alongside the frontier location for the rest. Collapsing the unidentified space into a small number of scenario bundles is avoided deliberately: bundles hide which single assumption is doing the work, and the one-parameter-at-a-time structure keeps every verdict traceable to the adjustment that could overturn it.

4.8 Data requirements and candidate implementations

The design runs on a minimum schema of five linked elements: longitudinal worker-employer earnings and separation records; establishment or firm accounts sufficient for quasi-rents (revenue or value added, wage bill, intermediate inputs, capital) and survival; predetermined shock exposure at the industry, region, or firm level; retention-program treatment, eligibility, and payment records if the causal tier is in scope; and the fiscal and insurance environment (UI receipt, taxes and transfers). Any environment containing these five elements can implement the design.

Three implementations are natural candidates. United States manufacturing under the trade shock links worker histories, establishment accounts, and trade exposure, and identifies the accounting tier at scale. The Paycheck Protection Program (PPP) links program loan records to worker-firm data and is the leading United States candidate for the causal tier— q and C —because delivery-channel variation is already documented [Granja et al., 2022, Autor et al., 2022]. German or Scandinavian administrative data pair matched employer-employee records and firm accounts with short-time-work programs whose eligibility and take-up rules provide cleaner policy variation than any United States retention episode [Giupponi et al., 2022, Cahuc, 2024]. Appendix E maps each environment’s variables to each estimand.

Table 1: The five empirical objects: definitions, identification, and remaining assumptions

Object	Empirical counterpart	Data and variation	Remaining assumption
u_i	Present-value earnings loss from matched separation event study, converted to welfare units net of taxes, transfers, and UI	Linked worker-firm panel; ex-ante separation and comparison definitions	Amenity value of the lost job; consumption smoothing beyond observed transfers
$\pi_i^{\text{priv}}, \pi_i^S$	Pre-shock establishment quasi-rent per worker (value added less wage bill less capital rental), allocated to matches; planner bridge $\hat{\pi}^S = \omega \hat{\pi}^{\text{priv}} + \dots$	Firm or establishment accounts linked to workforce; forward outcomes for validation only	Fixed-cost versus marginal-match allocation; planner weight ω on measured rents
$G(u, \pi)$	Joint distribution G_X of cross-fitted scores $(\hat{u}_i, \hat{\pi}_i)$ among pre-shock at-risk matches; empirical copula	Pre-shock panel; sample splitting for generated regressors	Latent copula set-identified around the score copula; residual within-cell heterogeneity
$p_z(u, \pi)$	Separation probability over $(\hat{u}, \hat{\pi})$ given predetermined exposure; heat map over $\hat{G}_X$	Exogenous or predetermined shock exposure; pretrend tests	Exposure exogeneity conditional on observables; within-cell selection bounded, not estimated
$q_{B,r,z}, C$	Complier distribution of saved matches and marginal cost per averted separation	Program eligibility discontinuities, timing, lender or administrative variation	External validity beyond compliers; job-year to separation conversion

Note: Objects one through four constitute the accounting tier and require no retention program; the fifth row adds the causal policy tier. Details for each row are in Sections 4.2–4.6.

5 Empirical Implementation and Planned Outputs

The implementation of Section 4 proceeds in one primary setting, in a fixed sequence, with each output answering a named theoretical object. This section states the sequence.

The analysis opens with the setting: the institutional environment, the shock and its timing, the sample of at-risk matches, the retention program if one is in scope, and the source of identifying variation for each tier. The second step estimates the two margins—the worker-side event-study losses and their welfare conversion, and the construction and validation of the firm-side quasi-rent proxy—delivering $\hat{u}_i$ and $\hat{\pi}_i$ for every at-risk match. The third step assembles the empirical triptych: the two estimated marginals and the estimated score-joint density G_X with the planner’s frontier drawn in, the direct empirical counterpart of Figure 1 and the paper’s central descriptive exhibit. The fourth step estimates shock selection: the fitted destruction kernel as a surface over the joint distribution, and the destroyed-surplus accounting $\bar{S}^D(z)$ against the efficient-selection benchmark. The fifth step, where the setting permits, estimates policy selection: the complier distribution of saved matches over the same joint distribution, the marginal cost per averted separation, and the comparison of the saved region with the destroyed region. The final step converts estimates into the partial welfare outputs of Section 4.7: rescue thresholds, cost thresholds, and the retention-versus-insurance verdict regions, reported as robust-region shares and frontier locations.

The planned headline exhibits follow the same sequence: the empirical triptych; the destruction-kernel surface over the joint distribution; the corresponding policy-response surface where estimable; a verdict-frontier figure over the unidentified-parameter axes; and two tables, one collecting joint-distribution and selection moments with uncertainty, the other collecting thresholds and robust-region shares. Together they populate every object that Sections 2 and 3 defined and no more; the structure of the completed analysis is fixed by the theory, and the data’s role is to fill it.

6 Applications and Interpretation

The estimates the design delivers would discipline three literatures, and the paper states the tests now rather than the conclusions.

6.1 Displacement estimates

The canonical displacement estimand is one coordinate of the destroyed-match value, sampled under a particular frame: tenure restrictions select on the worker coordinate and mass-layoff or closure conditioning selects on the firm’s private exit margin. The design turns the frame itself into an empirical object. With $\hat{G}$ and $\hat{p}_z$ in hand, the sign test is direct: is $\mathbb{E}[\hat{\pi}^S \mid \text{destroyed, frame}]$ below the at-risk average, and by how much does conditioning on tenure and distress move the joint distribution of the sampled matches relative to the population of destroyed matches? Whether measured worker losses overstate or understate destroyed joint surplus in the canonical frame is exactly what the estimates would settle, and the paper does not presume the answer.

6.2 Retention programs

Cost-per-job statistics for the Paycheck Protection Program, European short-time work, and similar programs become welfare-relevant only after two conversions the framework supplies: conditioning on the distribution of the matches the program actually saved, and expressing cost and value in the same units, per averted separation rather than per subsidized job-year. The causal tier of the design performs both conversions, and its output prices any given program’s marginal dollar rather than its average headline. Appendix C preserves, as an explicitly illustrative exercise, the break-even arithmetic showing how sensitive such conversions are to attachment, persistence, and unit choices.

6.3 Trade and persistent reallocation

Persistent shocks change both sides of the retention condition: the remaining continuation value behind $\bar{u}$ and $\bar{\pi}^S$ falls, and the duration over which any bridge subsidy must be paid lengthens. The design’s duration axis makes this a measured tradeoff rather than an assumption. The China trade shock is the natural pilot application—the worker coordinate is measured at scale, and Appendix D shows the firm coordinate is constructible from public industry data—but a public-data pilot cannot identify destroyed-match planner surplus, and the completed application awaits the linked implementation.

7 Conclusion

Worker losses do not, by themselves, price job retention. The displacement literature measures one coordinate of a two-coordinate object, under a sampling frame that conditions on the other coordinate’s private margin, and the policy debates that invoke those estimates ask questions—was the destruction inefficient, should the jobs have been saved, was insurance the right instrument—that the worker’s loss alone cannot answer. The paper’s framework states what would: the joint distribution of worker and planner-valued firm surplus among at-risk matches, the kernel describing which matches the shock destroyed, and, for policy evaluation, the kernel describing which matches the marginal program dollar saved, assembled into a shadow value and a retention-versus-insurance frontier.

The triptych organizes the measurement agenda those objects imply. Supply and demand identify the two marginal distributions, and two mature literatures have disciplined moments and proxies of each margin, though neither identifies the welfare coordinates of Section 3 exactly; the copula and the selection kernels are missing altogether. Section 4 specifies what linked employer-employee and firm-accounting data identify about each object—exactly, for the kernel-weighted means the welfare rule consumes; as bounds, for the latent dependence and within-cell selection—together with which conclusions follow without any retention pro-

gram and which additionally require program variation. The framework also fixes the honest reporting format for what will remain unidentified: verdict regions over the planner’s weight on measured rents, match duration, retention cost, and instrument-specific externalities, rather than categorical claims.

The payoff to completing the design is a set of answers the current evidence cannot supply: whether shocks destroy high- or low-total-surplus matches, whether programs save the matches shocks destroy or different ones, and where, in the space of costs and durations, retention beats insurance. Each is a question about the joint distribution and its kernels, and none can be answered one marginal at a time.

A Stylized Cases and the One-Dimensional Benchmark

This appendix reproduces the standard supply-and-demand diagrams behind Section 2.1, states the one-dimensional formalization of the selection loss behind Section 2.2, and records the macroeconomic corollary. Throughout the appendices, π denotes the planner-valued firm-side continuation surplus (π^S in the main text’s notation) except where the private object π^{priv} is explicitly flagged, $G(u, \pi)$ correspondingly denotes the planner-coordinate joint distribution (G^S in the main text, an innocuous change of variables under the monotone bridge of Section 3.2), and X denotes the retention externality X^R .

A.1 Flexible wages

When wages adjust freely, the labor market clears post-shock at a lower wage and lower employment. The destroyed matches are exactly those whose post-shock $u + \pi$ has reached zero on the margin and gone negative inframarginally. The marginal retained match has $u + \pi = 0$, and retaining any inframarginal destroyed match generates a strictly negative gross welfare gain. There is no efficiency case for retention: subsidizing employment above the new equilibrium retains matches with negative social surplus. Worker losses and social-

insurance motives still matter, but they enter the framework directly through the insurance margin, not through saved-job counts.

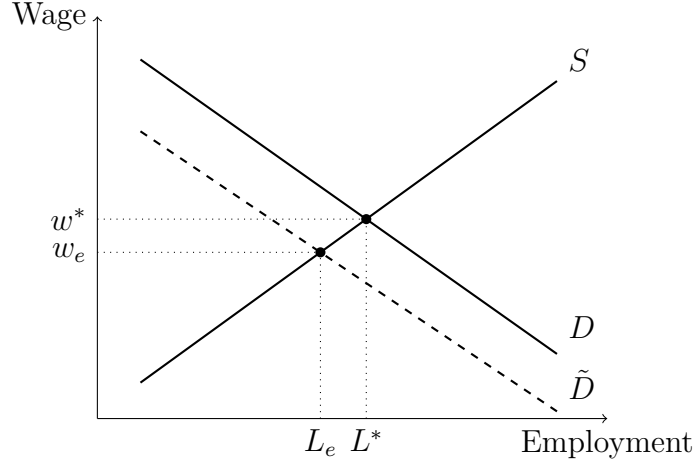


Figure 2: Flexible wages: the post-shock equilibrium is efficient.

Notes: Supply (S) and initial demand (D) intersect at the pre-shock equilibrium. A contraction to $\tilde{D}$ moves the equilibrium down along S . There is no welfare loss to shade: every destroyed match has $u + \pi = 0$ on the margin and $u + \pi < 0$ inframarginally.

A.2 Sticky wages

If wages do not adjust, the firm absorbs the shock by laying off matches whose marginal revenue product has fallen below the sticky wage. Employment falls more than it would under flexible wages: some matches with positive $u + \pi$ are inefficiently destroyed because the firm bears the entire shortfall through lower π . In this one-dimensional benchmark, the firm orders separations by π , and π is monotone in $u + \pi$ across matches. The inefficiently destroyed matches therefore all lie between the would-be flexible-wage equilibrium L_e and post-shock employment $\tilde{L}$. Their total surplus ranges from 0 at L_e to the wage wedge $\bar{w} - s(\tilde{L})$ at $\tilde{L}$. The welfare loss is a triangle whose area is second order in the wedge for small wage rigidity. The wedge between the sticky wage and post-shock marginal revenue product drives the firm's separation decision, but it is a transfer between worker and firm, not itself a social welfare loss.

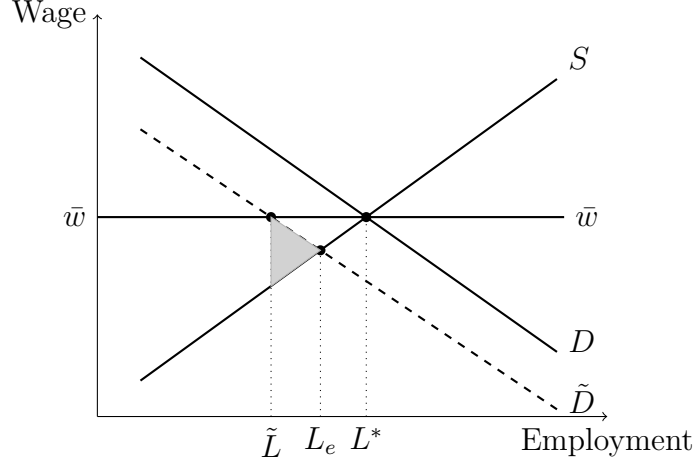


Figure 3: Sticky wages: matches in the shaded triangle have positive $u + \pi$ but are destroyed because the firm carries the full shortfall.

Notes: With $\bar{w}$ fixed, employment falls to $\tilde{L}$. Matches between L_e (where post-shock supply meets post-shock demand) and L^* have $u + \pi \leq 0$ and would have separated under flexible wages too. Matches in the shaded triangle, between $\tilde{L}$ and L_e , have positive $u + \pi$ but are destroyed because the firm cannot operate them at the sticky wage. Their total surplus ranges from 0 at L_e to the wage wedge $\bar{w} - s(\tilde{L})$ at $\tilde{L}$, giving a triangular welfare loss that is second order in the wedge for small wage rigidity.

A.3 Rationing and the formal selection loss

The sticky wage leaves more workers wanting jobs than there are jobs available, and the rationing logic follows [Luttmer \[2007\]](#). Workers in $[0, L^*]$ have positive surplus $u(L) = \bar{w} - s(L)$ at the sticky wage; only $\tilde{L}$ jobs survive the shock, so $m = L^* - \tilde{L}$ workers are without jobs. Under efficient rationing, the workers without jobs are those with the highest reservation wages, whose surplus is smallest; under random rationing, the workers without jobs have the population-mean surplus. The gross surplus destroyed by random rationing is $m \mathbb{E}[u]$, but the welfare loss relative to the efficient post-shock allocation subtracts what efficient rationing would have destroyed: the excess loss is $m(\mathbb{E}[u] - \mathbb{E}[u \mid \text{bottom } m])$.

The general statement replaces worker surplus with total match surplus and an arbitrary rationing rule. Let S_j denote the continuation surplus of match j among at-risk matches, and let $Q_S(\cdot)$ be the quantile function of S_j , ordered from low to high. For a shock z , let $D_j(z)$ indicate whether match j is destroyed, and let $q_z \equiv \Pr(D_j(z) = 1)$ be the separation rate. The efficient separation rule destroys the lowest-surplus matches first, with average

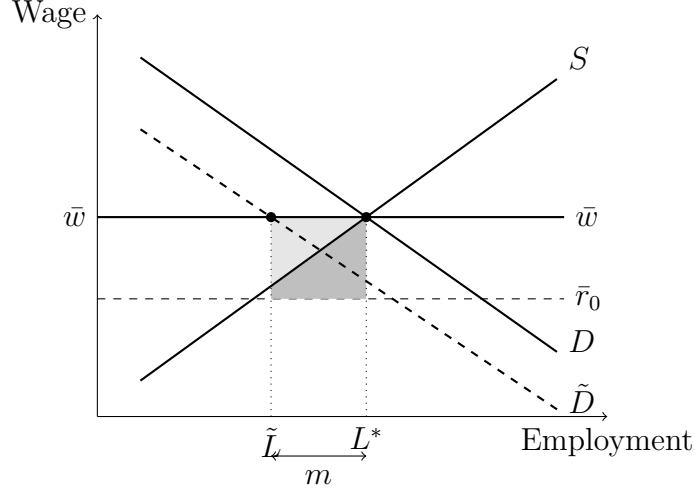


Figure 4: Sticky wages with random versus efficient rationing of $m = L^* - \tilde{L}$ workers.

Notes: Workers in $[0, L^*]$ have positive surplus $u(L) = \bar{w} - s(L)$ at the sticky wage; only $\tilde{L}$ jobs survive the shock, so $m = L^* - \tilde{L}$ workers are without jobs. Under random rationing, the workers without jobs have average surplus $\mathbb{E}[u] = \bar{w} - \bar{r}_0$, where $\bar{r}_0$ is the population-mean reservation wage; the gross surplus destroyed is the rectangle of width m and height $\mathbb{E}[u]$. Under efficient rationing, the workers without jobs are those with the highest reservation wages, on the rightmost $[\tilde{L}, L^*]$ of the supply curve; their lost surplus is the small triangle between $s(L)$ and $\bar{w}$. The welfare loss from inefficient rationing is the difference, $m(\mathbb{E}[u] - \mathbb{E}[u \mid \text{bottom } m])$.

nonfiscal loss per separation

$$\bar{S}_z^{\text{eff}} = \frac{1}{q_z} \int_0^{q_z} Q_S(v) dv.$$

Actual separation follows the shock-specific rule $D_j(z)$, with average destroyed surplus $\bar{S}_z^{\text{act}} = \mathbb{E}[S_j \mid D_j(z) = 1]$. The average excess loss per separation from inefficient rationing is

$$\Delta_z = \mathbb{E}[S_j \mid D_j(z) = 1] - \frac{1}{q_z} \int_0^{q_z} Q_S(v) dv, \quad (10)$$

and the aggregate excess loss is $q_z \Delta_z$, the formal version of equation (1). In the benchmark $S \sim U[0, w]$, aggregate destroyed surplus under random separation is $(w/2)q_z$, under efficient separation is $(w/2)q_z^2$, and the aggregate excess loss is $(w/2)q_z(1 - q_z)$. Random selection is the polar case; the substantive requirement for a first-order excess loss is only that the variables rationing continuation—cash reserves, bank credit, payroll obligations, bankruptcy timing, administrative eligibility, sectoral shutdown exposure, or the firm’s private continuation value—be imperfectly rank-ordered by total match surplus.

A.4 Macroeconomic corollary

The same distinction bears on business-cycle welfare accounting, provided gross and excess losses are kept apart. If destroyed jobs are inframarginal draws from $G(u, \pi)$, the gross surplus destroyed scales with a rectangle, $\mathbb{E}[u + \pi \mid \text{destroyed}, z] \cdot \Delta E$, rather than with a Harberger triangle around the marginal worker. The welfare loss relative to the efficient post-shock adjustment is smaller: it is the excess-selection component of equation (10), the gross rectangle less the surplus the efficient adjustment would have destroyed at the same separation rate. If separations are marginal or efficiently selected, welfare losses scale like a triangle; if separations are inframarginal and imperfectly targeted, the excess loss approaches the rectangle as selection worsens. The welfare cost of fluctuations therefore depends on the selection rule, not only on the employment loss. The flexible-wage, sticky-wage, and inframarginal-selection cases map naturally onto real-business-cycle, New Keynesian, and search-and-matching or heterogeneous-agent traditions [Lucas, 1987, Smets and Wouters, 2007, Shimer, 2005, Hagedorn and Manovskii, 2008, Krusell et al., 2009, Davis and von Wachter, 2011]; the macro job-rationing literature is especially close in spirit [Michaillat, 2012, Breza et al., 2021], and the reallocation debate over deep recessions [Caballero and Hammour, 1994, Barlevy, 2002, Foster et al., 2016] bears directly on whether productivity-shedding selection lowers the relevant continuation surplus. Different shocks generate different p_z over the same G , so the same headline employment loss can carry sharply different welfare cost; whether any particular recession destroyed high- or low-surplus matches is a hypothesis about $\bar{S}^D(z)$ that the design of Section 4 would test, not a fact the framework asserts.

B Formal Results and Proofs

This appendix states the general planner’s problem behind Section 3.4, the covariance form of the conditioning result, its bivariate-normal corollary, the observability decomposition,

the invariance properties of the shadow value, the notation table, and proofs.

B.1 The general joint planner's problem

Let g index policy cells (sector, location, wage, attachment) and t the macro state. The planner chooses retention spending $B_{g,t}$ for each cell and a single aggregate insurance generosity r . Because r is aggregate, the insurance margin is a common opportunity cost for all retention cells; cell heterogeneity enters Γ through the composition of the residual displaced pool, not through separate cell-specific insurance choices. Define the match-level shadow object

$$Y_{i,g,t}(r) = \Phi_{i,g,t}(r) + \frac{u_{i,g,t}(r) + \pi_{i,g,t} + X_{i,g,t}}{\lambda_t},$$

and the policy-marginal value of retention in cell (g, t) at spending B under shock z ,

$$P_{g,t}^m(B, r, z) = \frac{\int Y_{i,g,t}(r) q_{g,t,B,r,z}(u, \pi) dG_{g,t}(u, \pi)}{\int q_{g,t,B,r,z}(u, \pi) dG_{g,t}(u, \pi)},$$

where $q_{g,t,B,r,z}$ is the policy-marginal retention-response kernel of Section 3.2. Let $C_{g,t}(B, r, z) = (\partial J_{g,t} / \partial B_{g,t})^{-1}$ be the marginal public cost per averted separation in cell (g, t) , and let $\Gamma(B, r, z)$ be the aggregate Baily–Chetty marginal value of an additional public dollar of insurance, evaluated on the residual displaced pool.

Separability. The baseline theorem assumes that, conditional on retention spending B , insurance generosity r affects the welfare value of separations through $u_i(r)$, $\Phi_i(r)$, and the residual displaced pool, but not through the mechanical retention technology $J_{g,t}$ or the policy-response kernel $q_{g,t,B,r,z}$. Extensions in which r shifts J , C , or q through firm take-up, search behavior, or program targeting are accommodated by redefining Γ as the full marginal value of insurance inclusive of those cross-effects.

Theorem (joint retention–insurance optimum). Under standard differentiability, well-defined-kernel assumptions, and the separability above, the joint optimum has two related characterizations, depending on the financing environment.

Tax-financed planner. If the planner finances at marginal cost of public funds λ_t with no aggregate budget constraint, the optimum satisfies, for every cell (g, t) ,

$$\begin{aligned} B_{g,t}^* > 0 &\implies \frac{P_{g,t}^m(B^*, r^*, z)}{C_{g,t}(B^*, r^*, z)} = 1, \\ B_{g,t}^* = 0 &\implies \frac{P_{g,t}^m(B^*, r^*, z)}{C_{g,t}(B^*, r^*, z)} \leq 1, \end{aligned}$$

together with the insurance condition $\Gamma(B^*, r^*, z) = 1$ at an interior r^* .

Fixed-budget planner. If the planner has a fixed stabilization budget $\bar{Z}$ allocated between retention and insurance, the optimum satisfies, for every cell (g, t) ,

$$\begin{aligned} B_{g,t}^* > 0 &\implies \frac{P_{g,t}^m(B^*, r^*, z)}{C_{g,t}(B^*, r^*, z)} = \Gamma(B^*, r^*, z), \\ B_{g,t}^* = 0 &\implies \frac{P_{g,t}^m(B^*, r^*, z)}{C_{g,t}(B^*, r^*, z)} \leq \Gamma(B^*, r^*, z). \end{aligned}$$

The tax-financed problem yields two independent active-margin conditions; the fixed-budget problem yields an equalization condition across instruments. Section B.6 gives the proof. The fixed-budget condition rearranges, holding r at an exogenous $\bar{r}$, to the cost threshold (7) of the main text; that threshold is tabulated for the illustrative cells of Appendix C in the online appendix. The instrument-choice comparative statics recover those of Giupponi et al. [2022] and Liscow and Woolston [2017]: the planner subsidizes employers rather than employees when the policy-marginal social value of preservation, per dollar of retention spending, exceeds the marginal social value of insurance.

Implementing rule. With match-level observability of the firm's private continuation profit π_j^{priv} , the firm continues match j if and only if $\pi_j^{\text{priv}} + b_j \geq 0$, where b_j is any per-match subsidy. The minimum subsidy that preserves the match is $b_j^{\min} = \max\{0, -\pi_j^{\text{priv}}\}$, and the planner's value of preservation is $P_j^*(r^*) = \Phi_j(r^*) + (u_j(r^*) + \pi_j + X_j)/\lambda$ —the match-level evaluation of the shadow object $Y_{i,g,t}$ above, so $P_j^* \equiv Y_j$ —where π_j is the planner-valued firm-side surplus, not necessarily equal to π_j^{priv} . The optimal retention rule pays the subsidy if and only if the private shortfall is below the social value of preservation: save match j if and only if $b_j^{\min} \leq P_j^*(r^*)$. The rule presumes (a) b_j is a gross public-dollar subsidy with resource cost λb_j ; (b) firm-side transfers carry no independent distributional weight in the planner's objective; (c) π_j^{priv} is the firm's private continuation shortfall and π_j the planner-valued component, which may differ when product-market or monopsony rents do not reflect real social resources; and (d) the planner can target match j without inframarginal leakage. With only cell-level observability, the planner falls back on the cell-level condition $C_{g,t}(B_{g,t}^*) = P_{g,t}^m(B_{g,t}^*, r^*, z)$.

B.2 The covariance form of conditioning

Section 3.2 introduced the policy-marginal expected surplus as the kernel-weighted mean of $u + \pi$ over G . The comparative-static implication is exact.

Covariance identity. For any policy or cell h with kernel $q_h(u, \pi)$, the policy-marginal expected shadow value satisfies

$$\mu_h^q \equiv \frac{\mathbb{E}[Y q_h]}{\mathbb{E}[q_h]} = \mathbb{E}[Y] + \frac{\text{Cov}(Y, q_h)}{\mathbb{E}[q_h]}. \quad (11)$$

Consequently, for two policies or cells h_1 and h_2 with strictly different unconditional means $\mathbb{E}[Y_{h_1}] \neq \mathbb{E}[Y_{h_2}]$, the conditional ranking $\mu_{h_1}^q$ versus $\mu_{h_2}^q$ reverses the unconditional ranking if

and only if

$$(\mathbb{E}[Y_{h_1}] - \mathbb{E}[Y_{h_2}]) \left[\mathbb{E}[Y_{h_1}] - \mathbb{E}[Y_{h_2}] + \frac{\text{Cov}(Y_{h_1}, q_{h_1})}{\mathbb{E}[q_{h_1}]} - \frac{\text{Cov}(Y_{h_2}, q_{h_2})}{\mathbb{E}[q_{h_2}]} \right] < 0.$$

When the unconditional means coincide, conditioning can break the tie even though the product is zero. The decomposition is exact, follows directly from the definition of conditional expectation, and shows that the conditioning channel through which a shock or policy changes the welfare evaluation is the covariance between the kernel and the full shadow-value object Y , not the worker-firm correlation alone. In the flow-surplus benchmark of Section 2, where $\Phi = X = 0$, Y collapses to $(u + \pi)/\lambda$; because λ is common across compared objects, the scaling does not affect rankings.

Bivariate-normal corollary. A parametric corollary recovers the kind of correlation threshold the literature has used informally. Suppose (u, π) is bivariate normal with means μ_u, μ_π , standard deviations σ_u, σ_π , and correlation ρ . Suppose destruction selects on low firm surplus, $\pi < c$, and let $S = u + \pi$. Define the standardized cutoff

$$a_c = \frac{c - \mu_\pi}{\sigma_\pi}, \quad \eta(a_c) = \frac{\phi_{\mathcal{N}}(a_c)}{\Phi_{\mathcal{N}}(a_c)},$$

where $\phi_{\mathcal{N}}$ and $\Phi_{\mathcal{N}}$ are the standard normal density and cumulative distribution function (subscripted to avoid collision with the fiscal term Φ) and $\eta(a_c) > 0$ is the inverse Mills ratio at a_c . Then

$$\mathbb{E}[S \mid \pi < c] - \mathbb{E}[S] = -\eta(a_c) (\sigma_\pi + \rho \sigma_u),$$

and low- π selection raises the conditional mean above the unconditional mean if and only if $\rho < -\sigma_\pi/\sigma_u$: worker surplus must be sufficiently negatively correlated with firm surplus and sufficiently variable relative to it. The closed form is bivariate-normal; a Gaussian copula with non-normal marginals does not generally yield the same expression. Section B.6 gives the truncation derivation.

B.3 An observability decomposition

The policy-marginal expected shadow value admits an exact between-cell-plus-within-cell decomposition that organizes what public data can and cannot identify. Let c index observable public-data cells, and define the cell-level shadow-value mean, kernel mean, and population weight as

$$\bar{Y}_c = \mathbb{E}[Y_i \mid c], \quad \bar{q}_c = \mathbb{E}[q_i \mid c], \quad \psi_c = \Pr(c),$$

where ψ_c is the cell’s population weight (distinct from the planner weight ω of the main text). Then

$$\frac{\mathbb{E}[Y_i q_i]}{\mathbb{E}[q_i]} = \frac{\sum_c \psi_c \bar{q}_c \bar{Y}_c}{\sum_c \psi_c \bar{q}_c} + \frac{\sum_c \psi_c \text{Cov}(Y_i, q_i \mid c)}{\sum_c \psi_c \bar{q}_c}.$$

The first term is the between-cell component: identified only if ψ_c , $\bar{q}_c$, and $\bar{Y}_c$ are observed or credibly estimated. Public data and published estimates—displaced-worker losses by sector, age, and tenure; firm wage premia by sector and firm size; cell-level fiscal parameters—discipline $\bar{Y}_c$ under an explicit cell-level mapping. The second term is the within-cell residual: how much of the policy-marginal expected Y comes from sub-cell variation in q_i that the public data do not see. Linked employer-employee data, together with credible shock or policy variation that identifies q_i , estimate this residual directly; absent them, sensitivity over plausible within-cell rank correlations bounds the magnitude. The decomposition separates what the public-data pilots of Appendix D can deliver (the between-cell component) from what requires the linked design of Section 4 (the within-cell residual), and its empirical counterpart is the within- versus between-establishment variance decomposition of Section 4.4. The same decomposition applies one level down within the linked design itself: prediction cells play the role of public cells, the between component is what the score distribution G_X identifies, and the within-cell residual—dependence and selection operating inside prediction cells—is the component the design bounds rather than estimates.

B.4 What the shadow value depends on, and what it does not

The shadow value (4) is invariant to a range of modelling choices that matter elsewhere but not here. It does not depend on the mechanism generating wage stickiness (staggered bargaining, menu costs, implicit contracts, or monopsony power); the precise source of firm-side rents (matching surplus, rent-sharing, monopsony, or product-market rents); the functional form of the production function; or the financing instrument used to raise the subsidy, since λ absorbs that wedge. Each of these primitives affects the levels of $\bar{u}$, $\bar{\pi}$, $\bar{\Phi}$, or λ , but not their roles in the decomposition. The shadow value does depend on the joint distribution $G(u, \pi)$; on the destruction kernel p_z ; on the policy-response kernel $q_{B,r,z}$, which need not coincide with p_z ; on the insurance environment r through $\bar{u}(r)$ and $\bar{\Phi}(r)$ and the associated marginal value Γ ; and on the fiscal parameters Φ and λ . Different policies with the same reduced-form decomposition are welfare-equivalent at the margin, and different structural models that deliver the same reduced-form inputs generate the same shadow value; the policy-relevant primitives are a small set of reduced-form objects, in the tradition of [Baily \[1978\]](#), [Chetty \[2006\]](#), and [Saez \[2001\]](#). Distributional concerns are absorbed directly: weighting $\bar{u}$ by a welfare weight that rises as pre-separation income falls raises the shadow value in low-wage cells and can change rankings near any threshold; the framework accommodates such weights without modification.

B.5 Notation

Table 2 collects the paper’s notation. The shadow-value objects differ only in their conditioning: $\bar{S}^D(z)$ averages over the matches a shock destroys, and P^m conditions on the matches the marginal program dollar saves.

Table 2: Notation

Symbol	Definition
$u, u_i(r)$	Worker surplus: present-value welfare loss from separation, net of taxes, transfers, and insurance under environment r
π_i^{priv}	Firm’s private continuation profit: governs separation and subsidy take-up
π_i^S	Planner-valued firm-side continuation surplus; π in the appendices
$m(\cdot), \omega$	Planner bridge $\pi^S = m(\pi^{\text{priv}})$; scalar baseline $m(\pi^{\text{priv}}) = \omega \pi^{\text{priv}}$ with ω the planner weight on measured private operating surplus
$G(u, \pi^{\text{priv}})$	Joint distribution of worker loss and private firm surplus among at-risk matches; domain of behavior, kernels, and measurement
G^S	Planner-valued joint distribution of (u, π^S) induced from G by the bridge
G_X	Joint distribution of the cross-fitted scores $(\hat{u}_i, \hat{\pi}_i)$; the primary estimand of Section 4.4
$p_z(u, \pi)$	Destruction kernel: probability a match is destroyed under shock z
$q_{B,r,z}(u, \pi)$	Policy-response kernel: probability the marginal program dollar saves the match
q_z	Aggregate separation rate under shock z
S	Total match surplus, $u + \pi$
B	Retention-program spending; argument of $C(B)$ and $q_{B,r,z}$
z	Shock index
$\bar{S}^D(z)$	Destruction-weighted expected surplus of destroyed matches, equation (3)
$\bar{S}^m(B, r, z)$	Policy-marginal expected surplus of saved matches, equation (3)
X^R, X^I	Externalities mitigated by retention and by insurance or place-based support
$\Phi_i(r)$	Avoided net fiscal cost per averted separation
λ	Marginal cost of public funds
$P^m(B, r, z)$	Policy-marginal shadow value, equation (4)
$C(B)$	Marginal public cost per averted separation
C^*	Retention-versus-insurance cost threshold, equation (7)
$P_{g,t}^*, P_j^*$	Cell-level and match-level shadow values used in the appendices ($P_j^* \equiv Y_j$); equal to P^m when the program saves a representative draw of destroyed matches
Γ	Baily–Chetty marginal value of a public dollar of worker insurance
r	Insurance generosity
$b_i^{\min}$	Minimum bridge subsidy $\max\{0, -\pi_i^{\text{priv}}\}$, equation (8)
π_{rescue}^S	Firm-side surplus required to justify retention, equation (9)
$y_g, b_L(a_g), m(u_t)$	Calibration inputs (Appendix C): cell earnings, worker-loss archetype multiple, slack multiplier
ν_g	Job-year-to-separation conversion (Appendix C)
κ_g	Legacy reallocation share retained in sensitivity tables only (Appendix C)

B.6 Proofs

Proof of the joint retention–insurance optimum (Section B.1). The proof works at the margin. A marginal increase in retention spending in cell (g, t) averts $dJ_{g,t} = (\partial J_{g,t} / \partial B_{g,t}) dB_{g,t} = dB_{g,t} / C_{g,t}$ separations, each generating policy-marginal social value $\bar{S}_{g,t}^m(r) = \bar{u}_{g,t}(r) + \bar{\pi}_{g,t} + \bar{X}_{g,t}$ and avoided fiscal cost $\bar{\Phi}_{g,t}^m(r)$. The marginal welfare gain from those averted separations, net of the cost of public funds, is

$$dW_{g,t} = [\bar{S}_{g,t}^m(r) + \lambda_t \bar{\Phi}_{g,t}^m(r)] dJ_{g,t} - \lambda_t dB_{g,t}.$$

Using the policy-marginal shadow value $P_{g,t}^m(B, r, z) = \bar{\Phi}_{g,t}^m(r) + \bar{S}_{g,t}^m(r) / \lambda_t$, this rearranges to

$$dW_{g,t} = \lambda_t \left[\frac{P_{g,t}^m(B, r, z)}{C_{g,t}(B, r, z)} - 1 \right] dB_{g,t}.$$

The Kuhn-Tucker conditions for the tax-financed planner follow immediately: $B_{g,t}^* > 0$ requires $P_{g,t}^m / C_{g,t} = 1$, and $B_{g,t}^* = 0$ requires $P_{g,t}^m / C_{g,t} \leq 1$. The insurance condition follows from differentiating welfare in r : in the separable benchmark the residual displaced pool $N_D(r)$ accumulates welfare $h(r) - \lambda K(r)$ per worker, so $dW_r = N_D(r) [h'(r) - \lambda K'(r)] dr$, and an interior r^* requires $h'(r^*) = \lambda K'(r^*)$, equivalently $\Gamma(r^*) = 1$. In the general formulation $\Gamma(B, r, z)$ is defined as the full normalized marginal welfare return to one additional public dollar of insurance, inclusive of any cross-effects on J , C , or q flagged in the separability assumption; the first-order condition remains $\Gamma = 1$ at an interior optimum. Under a fixed stabilization budget $\bar{Z}$ that binds the financing constraint, the Lagrangian replaces λ_t by the budget multiplier; equating marginal welfare returns per public dollar across active instruments gives $P_{g,t}^m / C_{g,t} = \Gamma$ at active retention margins, with corner conditions analogous to the tax-financed case. The threshold (7) follows by substituting the components of $P_{g,t}^m$ into $P_{g,t}^m / C_{g,t} = \Gamma$ and rearranging for C^* at fixed $r = \bar{r}$; the rescue threshold (9) follows by instead solving for $\bar{\pi}^{S,m}$ at equality.

Proof of the covariance identity (Section B.2). For any random variable Y and weight function q with $\mathbb{E}[q] > 0$,

$$\frac{\mathbb{E}[Yq]}{\mathbb{E}[q]} = \frac{\mathbb{E}[Y]\mathbb{E}[q] + \text{Cov}(Y, q)}{\mathbb{E}[q]} = \mathbb{E}[Y] + \frac{\text{Cov}(Y, q)}{\mathbb{E}[q]},$$

which is (11). The ranking-flip condition follows by taking the difference $\mu_{h_1}^q - \mu_{h_2}^q$ and noting that the sign of the product of the unconditional difference and the conditional difference identifies whether the kernel-induced correction reverses the ranking.

Proof of the bivariate-normal corollary (Section B.2). Let (u, π) be bivariate normal with $\mathbb{E}[u] = \mathbb{E}[\pi] = 0$, $\text{Var}(u) = \sigma_u^2$, $\text{Var}(\pi) = \sigma_\pi^2$, and $\text{Corr}(u, \pi) = \rho$, so $\mathbb{E}[u \mid \pi] = (\rho\sigma_u/\sigma_\pi)\pi$. The standard truncated-normal mean gives $\mathbb{E}[\pi \mid \pi < c] = -\sigma_\pi \eta_c$ where $\eta_c = \phi_{\mathcal{N}}(c/\sigma_\pi)/\Phi_{\mathcal{N}}(c/\sigma_\pi) > 0$ is the inverse Mills ratio at the truncation. Then

$$\mathbb{E}[u \mid \pi < c] = (\rho\sigma_u/\sigma_\pi) \mathbb{E}[\pi \mid \pi < c] = -\rho\sigma_u \eta_c,$$

so $\mathbb{E}[S \mid \pi < c] = -\eta_c(\sigma_\pi + \rho\sigma_u)$, and $\mathbb{E}[S \mid \pi < c] - \mathbb{E}[S] = -\eta_c(\sigma_\pi + \rho\sigma_u)$ since $\mathbb{E}[S] = 0$. The truncation raises the conditional mean above the unconditional mean iff $\sigma_\pi + \rho\sigma_u < 0$, i.e., $\rho < -\sigma_\pi/\sigma_u$.

C An Illustrative Break-Even Calibration

This appendix preserves, as an explicitly illustrative exercise, the break-even calibration developed in an earlier version of this paper. Its purpose here is pedagogical: it shows how the decision rule of Section 3 behaves when populated with public data and assigned parameters, and how sensitive the resulting conclusions are to exactly the inputs the design of Section 4 would estimate. Nothing in it is an evaluation of the Paycheck Protection Program or of any other program. The sectoral cost figures are assigned scenarios anchored to a

single published aggregate range; the attachment classes that generate cross-cell variation are assigned by sector rather than estimated; the firm coordinate is set to zero in the baseline; and the joint distribution and selection kernels are assumed, not measured. The value of the exercise is that it makes the decision rule’s arithmetic transparent and shows which assignments the conclusions turn on.

C.1 Formula, units, and inputs

The exercise evaluates the cell-level shadow value

$$P_{g,t}^* = \bar{\Phi}_{g,t} + \frac{\bar{u}_{g,t} + \bar{\pi}_{g,t} + \bar{X}_{g,t}}{\lambda_t}, \quad (12)$$

against a marginal cost per job saved $C_{g,t}$, where π is planner-valued firm-side surplus and all terms are per averted separation in present value. Four inputs populate it, each with a different status. The marginal cost $C_{g,t}$ is anchored at the aggregate to [Autor et al. \[2022\]](#)’s \$169,000–\$258,000 range for the Paycheck Protection Program, which is the exercise’s only directly observed cost input; the sector-level cost values used below are assigned illustrative scenarios, not estimates, and the common-cost table described below shows the conclusions that survive without them. The worker-loss component $\bar{u}_{g,t} \equiv L_g = y_g \cdot b_L(a_g) \cdot m(u_t)$ is anchored to [Davis and von Wachter \[2011\]](#) and [Lachowska et al. \[2020\]](#) present-value displaced-worker earnings losses; the archetype multiple $b_L \in \{0.15, 0.60, 1.40\}$ is assigned by attachment class and the slack multiplier $m(u_t)$ rises in recessions (at the COVID April 2020 unemployment rate of 14.7 percent, $m = 2.0$). The attachment classes themselves are assigned by sector: published tenure and turnover statistics motivate the assignment, but no cell-level estimate underlies it, and this assignment is the single largest source of cross-cell variation in the results. The fiscal term is computable from tax and UI rules given that worker-loss path,

$$\Phi_g = \tau^{\text{er}} L_g + \theta^{\text{UI}} r^{\text{UI}} y_g \ell_g^{\text{UI}}, \quad (13)$$

with $\tau^{\text{er}} = 0.0765$ the employer-share payroll tax rate, $r^{\text{UI}} = 0.47$, $\theta^{\text{UI}} = 0.80$, and $\ell_g^{\text{UI}} = 26/52$. Both components are present values per averted separation, on the same horizon as the continuation terms they join in equation (12): employer-share payroll tax is levied on top of every dollar of gross earnings the separation forgoes, so the payroll component scales mechanically with the earnings-loss path that defines L_g , while the UI outlay is confined to the twenty-six-week covered spell. An earlier version of this paper instead applied the combined 15.3 percent payroll rate over the expected UI spell alone, which summed a sub-annual fiscal flow with multi-year present values inside P^* ; the present formulation makes the horizon uniform. Two netting conventions follow from using a gross-earnings anchor for the worker line, where Section 3.1 defines u net of taxes and transfers. Only the employer share of payroll taxes enters Φ_g : the employee share and income taxes on the forgone stream sit inside the gross-earnings worker line, where they are valued at $1/\lambda$ rather than dollar for dollar, understating P^* . And the UI outlay counted as a fiscal saving in Φ_g is not netted out of the worker line as a receipt, which overstates P^* by at most the UI component divided by λ —about \$13,000 in the air-transportation scenario below, small relative to the archetype-driven variation the exercise is designed to expose. The firm-side component is set to $\bar{\pi}_{g,t} = 0$ in the baseline; firm rents anchored to Kline et al. [2019] rent-sharing pass-through and Card et al. [2013] firm-FE premia enter only the joint sensitivity envelope reported in the online appendix, through a firm-rent multiple $b_M \in \{0.05, 0.20, 0.50\}$ assigned by attachment class (the firm-side analog of b_L), so the dollar component in the tables is $M_g^{\text{rent}} = y_g b_M$. The marginal cost of public funds is $\lambda = 1.3$.

Two unit conversions require flags. First, Autor et al. [2022] report cost per job-year, while the shadow value is per averted separation with full continuation value; the conversion scalar $\nu_g \in [0, 1]$ bridges the two, with baseline $\nu_g = 1$ defensible for that specific cost measure because its denominator is already the additional jobs preserved rather than all subsidized job-years, and $\nu_g = 0.5$ in the pessimistic direction; at the baseline the decision ratio $C/(\nu_g P^*)$ written in some table captions coincides with the C/P^* quoted in the text. Second,

an earlier version of this paper applied a sectoral reallocation haircut $(1 - \kappa_g)$ to continuation value in the pessimistic envelope. Because the worker-loss anchors are displaced-worker earnings losses measured against reemployment counterfactuals, worker-side reallocation is already inside u , and a separate haircut double counts it; the framework of Section 3 accordingly carries no reallocation term. The κ entries that remain in the sensitivity tables below are a legacy scenario retained for comparability with the earlier version and should be read as such. Table 3 summarizes how each primitive is measured.

C.2 Break-even arithmetic

The threshold condition $P_{g,t}^* \geq C_{g,t}$ rearranges to a transparent break-even statistic. Substituting $P^* = \bar{\Phi} + S/\lambda$ and solving for S at equality gives

$$S_{\min,g,t} = \lambda_t(C_{g,t} - \bar{\Phi}_{g,t}), \quad (14)$$

the continuation value the saved match must be worth, net of the avoided fiscal cost, to clear the break-even threshold. The corollary, expressed in earnings-equivalent units, is the number of years of annual earnings the match surplus must equal: $S_{\min,g,t}/y_g$. Reading the question this way recasts “is $C/P^* < 1$?” as “how valuable would the match have to be?”—the same inversion as the rescue threshold (9) of the main text. Each illustrative scenario below reports $S_{\min}$ and $S_{\min}/y_g$ alongside C/P^* so the arithmetic can be read in any of three equivalent currencies.

C.3 Illustrative scenarios

The scenarios span the range the assigned parameters generate for a temporary shock: a high-attachment scenario in which the assigned inputs clear the threshold (air transportation under PPP), and two low-attachment scenarios in which they fail it by an order of magnitude (retail trade and accommodation and food). The full seven-sector reference is in the online

Table 3: Measurement status of each empirical object.

Object	Status	Primary evidence	Sensitivity range	Key concern
$\Phi_{g,t}$ (avoided fiscal cost)	Rule-based given $\bar{u}_{g,t}$	UI component computed from replacement rate, spell duration, and take-up; payroll component is the employer-share rate applied to the present-value earnings-loss path, so both components are present values on the horizon of $\bar{u}_{g,t}$.	UI component $\pm 10\text{--}20\%$ (take-up); payroll component moves one-for-one with the worker-loss anchor.	Not independent of $\bar{u}_{g,t}$: a mis-assigned attachment class moves Φ and $\bar{u}$ together.
$\bar{u}_{g,t}$ (worker welfare loss)	Estimated from literature	Jacobson et al. [1993] , Davis and von Wachter [2011] , Lachowska et al. [2020] , Athey et al. [2026] ; cyclical and attachment heterogeneity.	Factor of 2 across cyclical states; factor of 3–5 across attachment.	Earnings loss is not identical to welfare loss; transfer offsets and consumption-equivalent adjustments matter.
$\bar{\pi}_{g,t}$ (match- specific value)	Calibrated	AKM firm FE premium $\times$ (1–rent-sharing) from Card et al. [2013] , Kline et al. [2019] . No-double-counting rule: excludes premia already in worker earnings.	Conservative: $\bar{\pi}=0$. Intermediate: firm-side rents net of pass-through. Aggressive: intermediate plus residual nonpecuniary match value.	Proxy may bundle rents, productivity, and amenities; double-counting risk with $\bar{u}$.
λ_t (MCPF)	Maintained assumption	Standard public-finance benchmark.	1.1–1.5.	Often treated too mechanically; common across cells in the baseline.
$J'_{g,t}(B)$ (jobs per dollar)	Estimated from literature	Autor et al. [2022] for PPP; sector/wage heterogeneity is the main gap.	\$169k–\$258k per job-year.	External validity across sectors, time, and policy environments.
ν_g (job-year to separation)	Maintained assumption	Fraction of the full per-separation continuation value realized by one marginal job-year of retention; not observed in PPP data.	0.5–1.0.	Converts cost-per-job-year statistic to per-separation value; favorable to retention at $\nu_g = 1$.
$G(u, \pi)$ (joint dis- tribution)	Maintained assumption	Marginals disciplined separately; Swedish closure evidence [Athey et al., 2026] is suggestive about heterogeneity. US linked employer-employee data required for direct estimation.	Tested qualitatively against the cross-context evidence review in the online appendix.	Central empirical unknown; direct US estimation requires LEHD access.
$p_z(u, \pi)$ (selection kernel)	Maintained assumption	Shock-to-quadrant taxonomy (online appendix); per-episode mass-layoff evidence.	Tested against the selection evidence reviewed in the online appendix.	Shock-specific conditioning is disciplined qualitatively, not identified.

appendix.

Table 4: Illustrative cases: break-even arithmetic for retention under the COVID PPP cost benchmark.

Cell	y_g	C_g	Φ_g	L_g	$S_{\min}$	$S_{\min}/y_g$ (yrs)	C/P^*	
							sectoral C	$C = \$200\text{k}$
Air transportation	\$90 k	\$169 k	\$36 k	\$253 k	\$173 k	1.9	0.73	0.87
Retail trade	\$34 k	\$200 k	\$7 k	\$10 k	\$251 k	7.5	13.47	13.47
Accommodation & food	\$22 k	\$170 k	\$5 k	\$7 k	\$215 k	9.6	17.11	20.13

Notes: Three cells from the seven-sector reference table in the online appendix. y_g : annual earnings from QCEW 2019. C_g : PPP marginal cost per averted separation, obtained from Autor et al. [2022]’s cost per job-year at the baseline conversion $\nu_g = 1$ and assigned within the \$169k–\$258k aggregate range. Φ_g : avoided UI outlay plus employer-share payroll tax on the present-value earnings-loss path, equation (13). $L_g = y_g \cdot b_L(a_g) \cdot m$ is the worker-loss component of continuation surplus (baseline $\pi = 0$: firm rents reported as sensitivity in the online appendix). $S_{\min} = \lambda(C - \Phi)$ with $\lambda = 1.3$ is the continuation value the match must equal to clear $P^* \geq C$; $S_{\min}/y_g$ is the same threshold expressed in years of annual earnings. The two C/P^* columns use sector-specific C_g (left) and a uniform $C = \$200\text{k}$ across cells (right) to show that the cross-cell ranking and order-of-magnitude spread come from heterogeneity in P_g^* , not from the sectoral cost assignment. Air transportation clears the break-even threshold; Retail and Accommodation require 7–10 years of annual earnings to clear it, an order of magnitude above the calibrated L_g .

Air transportation. Air transportation is the cleanest temporary high-attachment configuration in the COVID episode: routes were idled by demand collapse rather than by productivity decline, and large firms with strong attachment faced a liquidity rather than a viability problem. Under the assigned inputs, the break-even threshold $S_{\min} \approx \$173\text{k}$ requires the continuation value to be worth about 1.9 years of annual earnings, inside the displaced-worker literature’s present-value estimates for high-tenure workers, and the scenario’s $C/P^* = 0.73$ survives replacing the assigned sectoral cost with a uniform $C = \$200\text{k}$ (ratio 0.87). The scenario illustrates the mechanism—expensive retention can clear a break-even threshold when the shock is temporary and the saved matches are high-attachment—under assignments the linked-data design would replace with estimates.

Retail and accommodation and food. The two low-attachment scenarios fail the break-even test by an order of magnitude under the same assignments: accommodation and food

would need a continuation value of about 9.6 years of annual earnings to clear the threshold, retail about 7.5 years, against calibrated worker-loss components of roughly \$10,000 or less. The reading is about instrument choice under the assigned inputs, not about program failure: where calibrated continuation value is small, helping the worker through insurance or transfers is unchanged in importance while the case for preserving the specific match weakens.

What the contrast shows. The two poles bracket the sensitivity of the decision rule to attachment and persistence: the same aggregate cost benchmark maps into a ratio below one or above nine depending on assignments that public data cannot verify at the cell level. That sensitivity is the exercise’s substantive point, and it is the case for estimating the joint distribution and kernels rather than assigning them. Appendix [D.1](#) presents the permanent-shock counterpart as a public-data pilot.

C.4 Sensitivity

Table [5](#) reports a one-at-a-time sensitivity for the air transportation and retail scenarios, varying each of $\{\lambda, \nu, \kappa, b_L, m\}$ around the baseline with the others held fixed, plus a firm-rents-on swing (b_M). Air transportation crosses the break-even threshold under two single-parameter swings— $\nu_g = 0.5$ and a halving of the worker-loss anchor—so the clears-the-threshold scenario is sensitive to the job-year-to-separation conversion and to the worker-loss anchor. Retail remains $C/P^* > 10$ under every plausible single-parameter swing. The κ row is the legacy reallocation scenario discussed in Section [C.1](#) and is retained for comparability only. The joint envelope that bundles the swings together is reported in the online appendix; the one-at-a-time table is the more informative object because it keeps every movement traceable to a single assignment.

Table 5: One-at-a-time sensitivity of $C/(\nu_g P^*)$ for Air Transportation (yes-case) and Retail trade (no-case).

Swing	Air transportation	Retail trade
baseline	0.73	13.47
$\lambda = 1.5$	0.83	14.48
$\lambda = 1.05$	0.61	11.98
$\nu_g = 0.5$	1.46	26.94
legacy reallocation scenario ($1-\kappa_g$)	0.73	15.73
half worker loss ($b_L \times 0.5$)	1.36	18.90
1.5 $\times$ worker loss	0.50	10.47
firm rents on ($b_M = \text{AKM/Kline}$)	0.56	11.47
deep-recession slack ($m \times 1.5$)	0.50	10.47

Notes: Each row changes one parameter from the baseline (others held at baseline); the entry is $C/(\nu_g P^*)$ under that single swing. Air Transportation crosses the break-even threshold $C/P^* = 1$ under two single-parameter swings ($\nu_g = 0.5$ and half worker loss), so the yes-case is sensitive to the job-year-to-separation conversion and to the worker-loss anchor. Retail trade remains $C/P^* > 10$ under every plausible single-parameter swing: the no-case is robust to plausible single-parameter perturbations. For the joint envelope that bundles these swings together see the robustness table in the online appendix.

C.5 Maintained assumptions

The calibration rests on a small set of maintained assumptions; this subsection collects them, states which results depend on each, and gives the direction of bias where it can be signed.

The first is the job-year-to-separation conversion $\nu_g = 1$. It is load-bearing for the exercise’s only clears-the-threshold scenario: at $\nu_g = 0.5$ the air-transportation ratio moves from 0.73 to 1.46. Its defense is specific to the [Autor et al. \[2022\]](#) cost measure, whose denominator is already the additional jobs preserved by the program; for programs whose cost estimates count subsidized job-years, $\nu_g = 1$ would not be defensible and the threshold would be correspondingly harder to clear.

The second is the working approximation $q_{B,r,z} \approx p_z$: the marginal program dollar saves a representative draw from the matches the shock destroys. For PPP the direct evidence runs against this: [Granja et al. \[2022\]](#) show that funds initially flowed disproportionately to firms with stronger banking relationships and to less-affected regions—selection on the firm coordinate, which is exactly the failure mode the covariance identity of Appendix B.2 formalizes. The approximation overstates how closely the saved matches track the destroyed

ones; the causal tier of Section 4.6 replaces it wherever program variation permits.

The third is the separability of retention and insurance: generosity r shifts $u(r)$, $\Phi(r)$, and the residual displaced pool, but not the retention technology or the response kernel. Short-time-work evidence suggests this is not innocuous, because take-up of employer-side instruments responds to the generosity of the worker-side alternative [Giupponi et al., 2022]; the general formulation of Appendix B.1 absorbs such cross-effects by redefining Γ , but the calibration does not quantify them.

The fourth is the conservative firm coordinate, $\bar{\pi} = 0$, which shuts down the paper’s central object within the exercise. The direction is unambiguous: P^* is understated, so the fails-the-threshold scenarios are strengthened, while the air-transportation scenario clears without firm rents.

The fifth is the archetype mapping itself: the cross-cell variation in P^* is generated by assigning b_L by attachment class and multiplying by cell wages, not by estimating anything at the cell level. This is the precise sense in which the exercise is a worked example rather than an estimate. The common-cost column of Table 4 and the uniform- C exercise reported in the online appendix show the cross-cell ranking survives the cost assignment, so the dispersion in C/P^* reflects the assigned heterogeneity in P_g^* ; what no public-data exercise delivers is the within-cell joint distribution, which is the object of Section 4.4.

C.6 Limits

The identification gap that remains within public-data constraints is the joint distribution $G(u, \pi)$ together with the two kernels; each requires matched employer-employee microdata. A stylized exercise reported in the online appendix isolates the role of the copula by holding the marginals fixed and varying the rank correlation between u and π : under endogenous-decline selection, raising the rank correlation from 0 to 0.7 lowers the air-transportation shadow value from roughly \$211,000 to \$166,000 with both marginals unchanged—at the assigned sectoral cost of \$169,000, enough on its own to move the scenario from clearing the

break-even threshold to failing it. The copula is not a nuisance parameter. In real time, sector, payroll, firm age, and tenure proxies are observable quickly in program administration, and QCEW, QWI, BDS, and LAUS aggregates can discipline a between-cell calibration within weeks or months; what cannot be observed in real time is the continuation-value margin itself. A practical substitute is an attachment classification by sector, motivated by QWI turnover and tenure statistics but assigned rather than estimated (tabulated in the online appendix); the linked-data design is what would replace the assignment with measurement.

D Public-Data Pilots

This appendix reports two pilot exercises using public data. Their common purpose is feasibility: to show that the accounting objects of Section 3 can be constructed at the industry and sector level from public sources, and to locate precisely where public data stop and linked microdata become necessary. Neither pilot identifies destroyed-match planner surplus, and neither supports a policy verdict.

D.1 The China trade shock: constructing the firm coordinate from public data

The trade-adjustment literature has measured both coordinates of the destroyed-match value—in separate papers, for separate purposes, without adding them. On the worker side, [Autor et al. \[2014\]](#) find cumulative 1992–2007 earnings reductions of 46 percent of initial annual earnings for manufacturing workers at the 75th relative to the 25th percentile of exposure to Chinese import competition. On the firm side, [Autor et al. \[2020\]](#) document falling sales and profitability at import-exposed firms; [Pierce and Schott \[2016\]](#) show manufacturing employment fell 18 percent between 2001 and 2007, concentrated in industries facing the largest reductions in tariff uncertainty; and [Fort et al. \[2018\]](#) show that more than

80 percent of the long-run decline in manufacturing employment occurred through establishment deaths. The pilot asks whether the firm coordinate can be constructed at scale and related to exposure using public data alone.

It can, at the industry level. The NBER-CES Manufacturing Industry Database reports value added, payroll, employment, and capital stocks by six-digit NAICS industry from 1958 through 2018, so an industry-level quasi-rent per worker—value added net of payroll and a capital rental charge, per employee—is directly computable for exposed and non-exposed industries over the shock period:

$$QR_{i,t} = \frac{VA_{i,t} - W_{i,t} - r^K K_{i,t}}{E_{i,t}}, \quad (15)$$

where $VA_{i,t}$ is value added, $W_{i,t}$ total payroll, $K_{i,t}$ the real capital stock, r^K a user cost of capital, and $E_{i,t}$ employment in industry i and year t . The quasi-rent is the flow counterpart of the firm-side coordinate: what the industry’s matches generate above the cost of labor and capital, per job. Figure 5 plots the employment-weighted quasi-rent per worker for industries in the top and bottom quartiles of Chinese import exposure between 1990 and 2016, alongside each group’s employment. Two facts stand out. First, the exposed quartile is manufacturing’s low-quasi-rent segment: over 2001–2007 its quasi-rents averaged roughly \$84,000 per worker per year in 2018 dollars, less than half the roughly \$191,000 of the least-exposed quartile, with payroll per worker of about \$48,000 against \$57,000. Second, destruction concentrated exactly there: exposed-quartile employment fell 44 percent between 2001 and 2011, against 16 percent in the least-exposed quartile. The survivor series also behaves as selection on the firm coordinate predicts: exposed-quartile quasi-rents dip at accession and then rise as employment collapses, the mechanical signature of the lowest- π matches exiting first.

Three features of the exercise mark exactly where the public data stop. First, the industry-year quasi-rent is an average across surviving establishments, not the quasi-rent

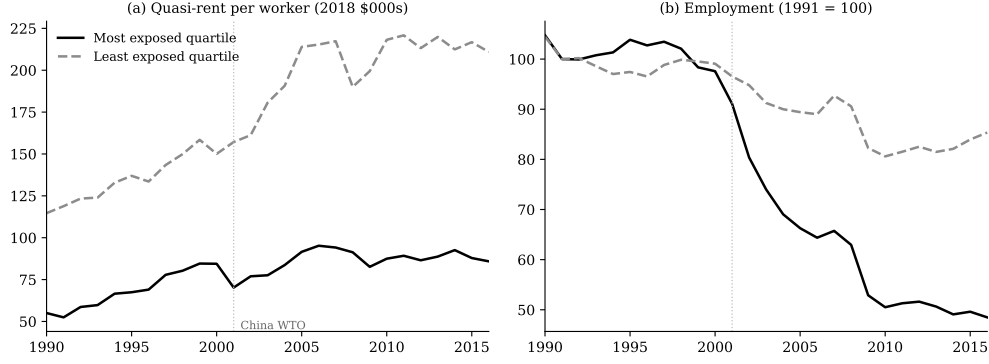


Figure 5: Chinese import competition and manufacturing’s low-quasi-rent segment: a public-data pilot of the firm coordinate.

Note: Quasi-rents per worker from equation (15), NBER-CES Manufacturing Industry Database (1990–2016; the capital series ends in 2016), employment-weighted within exposure group, deflated to 2018 dollars with the CPI-U. Exposure is the 1991–2007 change in Chinese import penetration at the industry level [Acemoglu et al., 2016], apportioned to NAICS-2012; quartiles are employment-weighted using 1991 employment, so each holds one quarter of 1991 manufacturing employment. The capital rental uses a 10 percent user cost; a gross-of-capital variant raises both series without changing the comparison.

of the marginal destroyed match; because dying plants exit the aggregate before they stop being counted, the surviving-plant average overstates the firm-side surplus of the matches actually destroyed. Second, the capital rental deduction uses a common user cost across industries; a gross-of-capital variant raises measured quasi-rents everywhere without changing the comparison. Third, value added embeds product-market rents that the planner may not count at full value, per the no-double-counting rule of Section 3.1. On all three margins the measured QR is an upper bound on the planner-valued π of the destroyed matches—and none of the three can be resolved without establishment-level linked data.

Table 6 assembles the accounting in the format of Appendix C, as an illustration of the rescue-threshold logic under stated assignments rather than as an evaluation. The worker side takes $y_g \approx \$48,000$ (exposed-quartile payroll per worker in 2000, 2018 dollars) with the high-attachment anchor $b_L = 1.40$ and $m = 1.0$; the fiscal term follows equation (13), giving $\Phi_g \approx \$14,000$; and with $\bar{\pi} = 0$ the shadow value is $P^* \approx \$66,000$ against a \$200,000 benchmark retention cost. Inverting at that cost, retention would have required planner-valued firm-side continuation value of roughly \$174,000 per worker—3.6 years of the cell’s annual earnings—which is the pilot’s version of the rescue threshold (9). Under a persistent

shock the private implementation of Section 3.5 also prices what preservation would have required on the cost side: a bridge subsidy $b^{\min} = -\pi^{\text{priv}}$ per period for as long as the decline persisted, on top of program cost, for matches sitting at the firm’s exit margin by revealed preference. Both calculations are conditional on the assigned inputs and on the surviving-plant average standing in for destroyed matches; replacing those substitutions with establishment-level estimates is precisely what the linked design does.

Table 6: The China-shock cell: the threshold arithmetic of structural decline.

Case	y_g	L_g	Φ_g	P_g^*	$C^*(\Gamma=1)$	$C^*(\Gamma=0.7)$
China-shock exposed manufacturing	\$48 k	\$68 k	\$14 k	\$66 k	\$66 k	\$95 k

Notes: The cell is the top quartile of Chinese import-penetration exposure (Acemoglu et al., 2016 industry measure, 1991–2007, apportioned to NAICS-2012), holding 25 percent of 1991 manufacturing employment across 92 six-digit industries. y_g : payroll per worker in the exposed quartile in 2000, in 2018 dollars (NBER-CES). $b_L = 1.40$ (Davis and von Wachter [2011] high-tenure anchor), $m = 1.0$ (structural decline is persistent, not a temporary aggregate shock), $b_M = 0$, so $L_g = y_g b_L m$. $P^* = \Phi + L/\lambda$ with $\lambda = 1.3$; $C^* = (L + \lambda\Phi)/(\lambda\Gamma)$ from equation (7). Memo: for retention to be preferred over insurance at the \$200,000 PPP-benchmark cost, the destroyed matches’ planner-valued continuation surplus $\bar{\pi}$ would need to be \$174 k per worker in present value (3.6 years of annual earnings); under endogenous decline the destroyed matches sat at the firm’s exit margin ($\pi^{\text{priv}} \leq 0$). Destruction was concentrated in manufacturing’s low-firm-value segment: exposed-quartile quasi-rents per worker (an annual flow) averaged \$84 k over 2001–2007, less than half the \$191 k of the least-exposed quartile, and exposed-quartile employment fell 44 percent between 2001 and 2011 (least-exposed: 16 percent). See Appendix D.1.

A full design in this setting is stated in Section 4 terms. Exposure enters continuously rather than by quartile, using the industry-level import-penetration change instrumented by other high-income countries’ imports from China [Autor et al., 2013, Acemoglu et al., 2016]; establishment-level quasi-rents replace industry survivor averages; an event-study specification traces both worker and establishment outcomes around exposure with pretrend tests; and the destruction kernel is estimated over the pre-shock joint distribution of predicted worker losses and establishment quasi-rents. The public pilot demonstrates the inputs exist; the linked implementation is what turns them into the paper’s estimands.

D.2 Public sector-level cells: the between-cell component

The second pilot is the sector-cell apparatus of Appendix C itself, read as a demonstration of what public aggregates identify. QCEW provides cell wages and employment; QWI provides turnover and tenure proxies that motivate attachment classifications; BDS provides establishment entry and exit; LAUS provides slack. In the language of the observability decomposition of Appendix B.3, these sources discipline the between-cell component of the policy-marginal shadow value: cell-level means of worker losses (through published displacement estimates mapped to cells), fiscal parameters, and costs. What they cannot identify is any within-cell object: tenure at the match level, the copula linking worker and firm surplus, or within-cell policy selection. The between-cell component is a genuine input to the completed analysis—it is the component against which the linked estimates’ within-cell residual will be compared—but it cannot substitute for the linked design, and the calibration’s sensitivity to assigned attachment classes is the measure of how much is left on the table.

E Data Designs and Access Paths

This appendix describes four candidate data environments for the design of Section 4, mapping each to the five estimands. The descriptions are deliberately brief; their purpose is to let a reader with access to one of these environments, or knowledge of a better one, judge feasibility quickly.

United States manufacturing under the trade shock (FSRDC). The Longitudinal Employer-Household Dynamics data provide worker-employer earnings histories and separations (u , and the worker score $\hat{u}_i$); the Longitudinal Business Database provides establishment survival; the Annual Survey of Manufactures and Census of Manufactures provide value added, payroll, materials, and capital for establishment quasi-rents (π^{priv}); industry import-

penetration changes with the other-country instrument provide predetermined exposure (p_z) [Autor et al., 2013, Acemoglu et al., 2016]; and UI wage records inside LEHD support the fiscal term. No retention program operates in this setting, so the causal tier is out of scope and the deliverables are the accounting tier plus thresholds. Access runs through the Federal Statistical Research Data Center system, and the linkage pattern (LEHD-LBD-ASM/CMF) is established in prior work.

The Paycheck Protection Program (SBA-Census linkage). SBA loan-level records identify treatment, amount, timing, and lender; linked to LEHD and the LBD they support all five objects, and the delivery-channel variation documented by Granja et al. [2022]—bank relationships, timing of tranche exhaustion, size-eligibility rules—provides the response-kernel identification of Section 4.6. This is the rare United States environment in which q and C are identified for a large retention program [Autor et al., 2022]. The limitation is the firm-accounts side: quasi-rents are directly constructible for the covered manufacturing subset, and require revenue-based proxies elsewhere.

Germany: LIAB and short-time work (IAB-FDZ). The IAB’s linked employer-employee data (LIAB) pair worker histories with establishment surveys, and Germany’s short-time-work program (Kurzarbeit) has eligibility and take-up rules, scale, and multiple episodes (2008–09, 2020) that make it the cleanest retention-policy variation available anywhere (q, C) [Giupponi et al., 2022, Cahuc, 2024]. German displacement designs are mature [Schmieder et al., 2023, Fackler et al., 2021], supporting $\hat{u}_i$; the establishment survey supports revenue- and wage-bill-based surplus proxies, though capital measures are weaker than in United States manufacturing data, so the π construction leans on the bounding strategy of Section 4.3. Access runs through the FDZ visiting-researcher system.

Scandinavian registers. Swedish, Danish, and Norwegian register data link workers, firms, and firm balance sheets population-wide, with the Swedish data already supporting

the sorting and within-establishment heterogeneity evidence the paper builds on [Bonhomme et al., 2019, Athey et al., 2026]. Firm accounts include capital, supporting full quasi-rent construction (π^{priv}); mass-layoff notification systems provide sharply defined destruction events (p_z); and the 2020 short-time-work schemes provide retention-policy variation (q, C). These environments dominate on data quality; the tradeoff is institutional distance from the United States policy debate the paper addresses.

The mapping across environments is deliberately redundant: any one of the four implements the accounting tier, and three of the four contain a retention program for the causal tier.

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